



PT SAPTAINDRA SEJATI

ENGINE ASSY

RECEIVING INSPECTION

CHECKSHEET

KOMATSU ENGINE 12V140E-3 SERIES

MACHINE MODEL

HD785-7

RO NO :

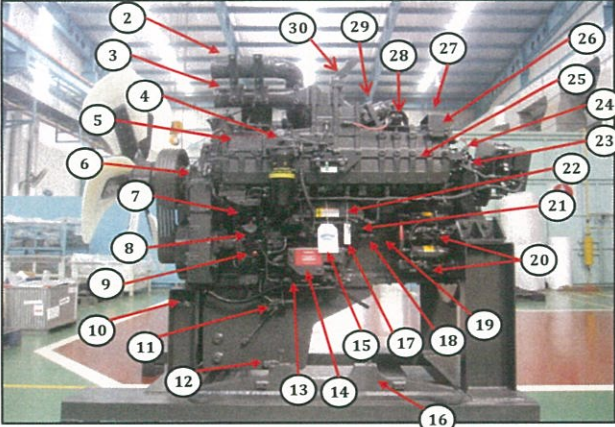
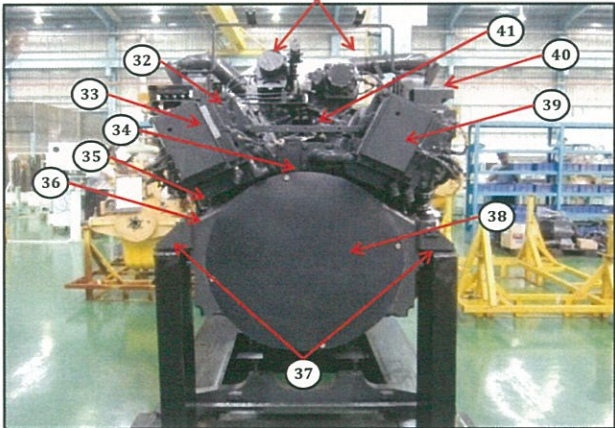
ENGINE SECTION

RECEIVING INSPECTION SHEET ENGINE

MACHINE MODEL	HD785-7	ENGINE PART NO.		DOCUMENT NO.	RC/QR01/EG/01/01
ENGINE MODEL	SAA12V140E-3	WO ORDER NO.		PUBL. DATE	
ENGINE NO.		RECEIVING DATE		REVISION NO.	01
CUSTOMER NAME				PAGE	1 OF 2

BELOW ARE THE ITEMS RELATED TO DELIVERY INSPECTION AND WHILE YOU DOING THIS INSPECTION JOB, PLEASE REFERRING TO THE REMANUFACTURED COMPONENTS - STANDARD CONFIGURATION.

✓ : GOOD CONDITION X : BAD CONDITION ⊗ : NONE

SKETCH DRAWING OR PHOTOES	NO.	ITEM	INSPECTION CONDITION
Left Side View 	1	Painting (Y/B)	
	2	Tube Air Intake Connection LH	
	3	Air Connector LH	
	4	Fuel Filter Assy LH	
	5	Engine Name Plate	
	6	Tube, Breather With Hose LH	
	7	Fuel Supply Pump Assy LH (P/N : 6219-71-1110)	
	8	Tube Oil Filler Cap	
	9	Gauge Oil (STD : H+ 0~10 mm)	
	10	Engine Stand (Original / Std)	
	11	Oil Level Sensor Assy	
	12	Drain Oil Plug	
	13	Priming Pump Assy With Cover LH	
	14	Warning Sticker Oil Level	
	15	Bracket Mounting	
	16	Quality passed Inspection Tag	
	17	Caution Tag	
	18	Oil Temperature Sensor	
	19	Oil Pressure Sensor	
	20	Starting Motor Assy (7.5 KW) (P/N : 600-813-7542)	
		Upper S/N :	
		Lower S/N :	
	21	Fuel Pressure Sensor With Limiter Assy LH	
	22	Common Rail Assy	
	23	Boost Pressure Sensor	
	24	Intake Temperature Sensor	
	25	Air Intake Manifold LH	
	26	Exhaust Temperature Sensor (4 pcs) With Bracket	
	27	Customer Responsibilities Sticker	
	28	Heater Switch Assy With Cover LH	
	29	Exhaust Brake Assy LH	
	30	Bracket Exhaust Brake Mounting	
Rear Side View 	31	Turbocharger Assy RH P/N : 6505-67-5030	
		S/N :	
		Turbocharger Assy LH P/N : 6505-67-5040	
		S/N :	
	32	Heat Tab position	
	33	Engine Controller Assy LH	
		S/N : ESN :	
		P/N Assy :	
	34	Caution Flywheel Sticker	
	35	Pick Up Revolution Sensor	
	36	Flywheel Housing	
	37	Rear Bracket Mounting (2 pcs)	
	38	Flywheel (P/N : 6219-31-4110)	
	39	Engine Controller Assy RH	
		S/N : ESN :	
		P/N Assy :	
	40	Stiker Attention Installation Report	
	41	Exhaust Manifold Assy RH & LH	

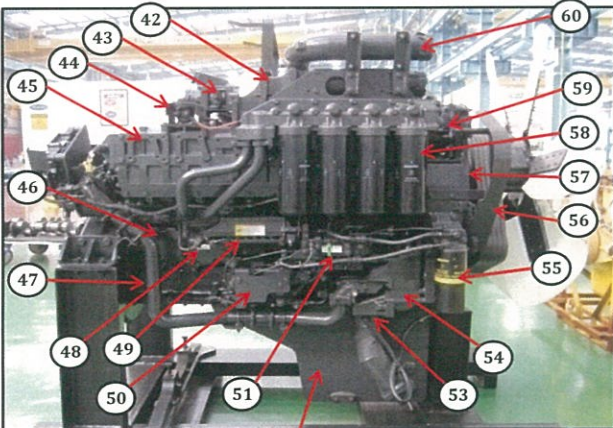
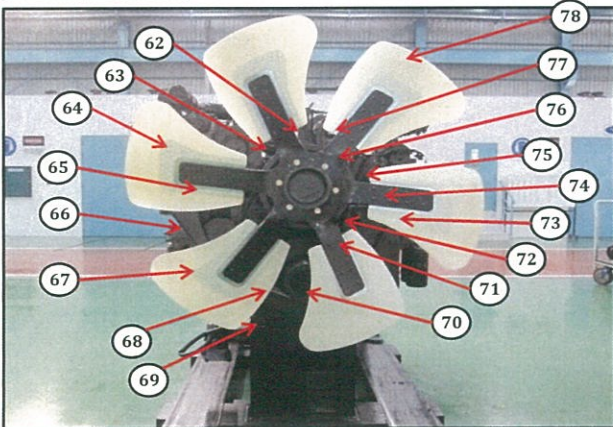
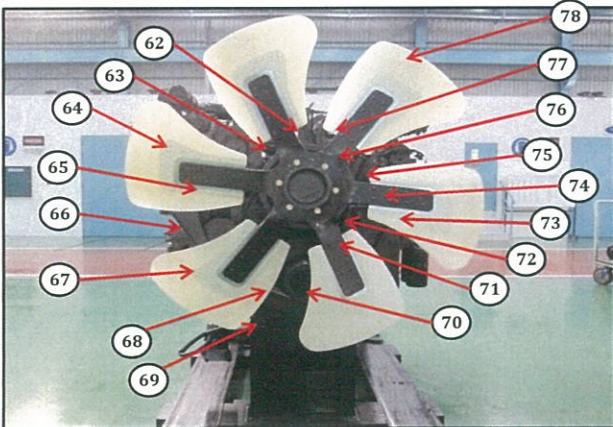
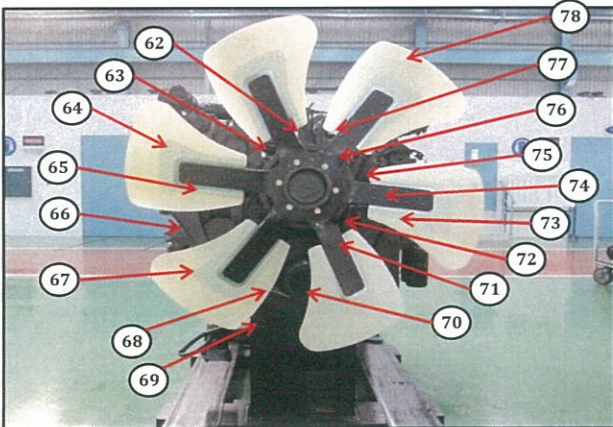
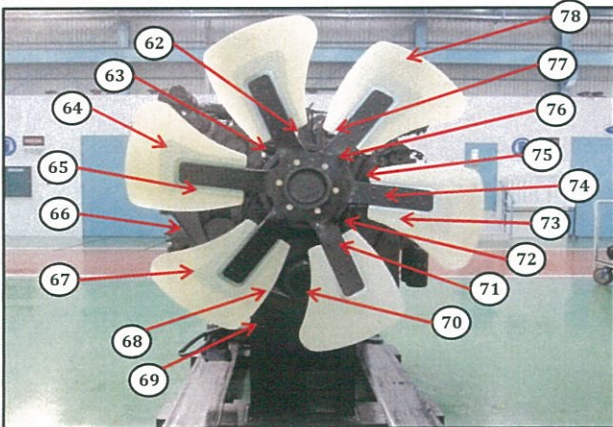
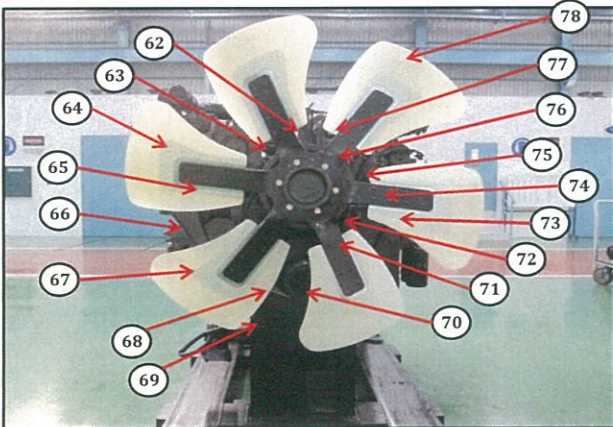
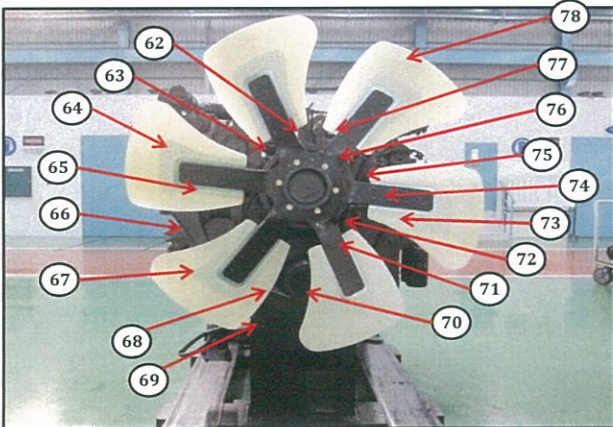
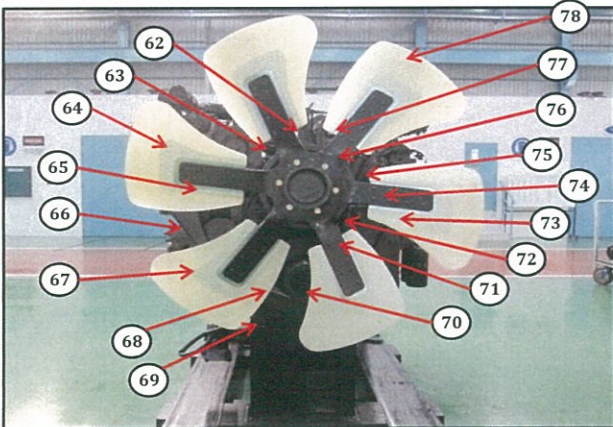
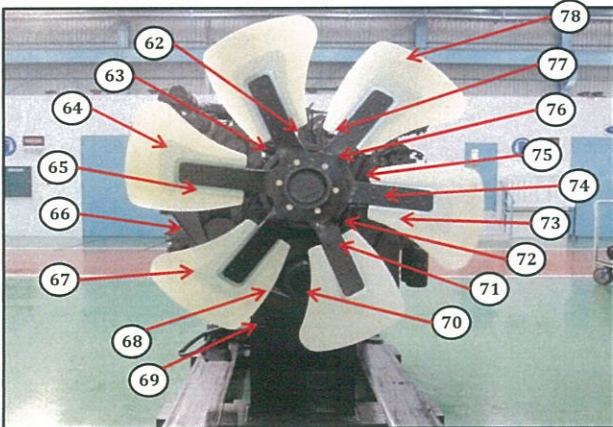
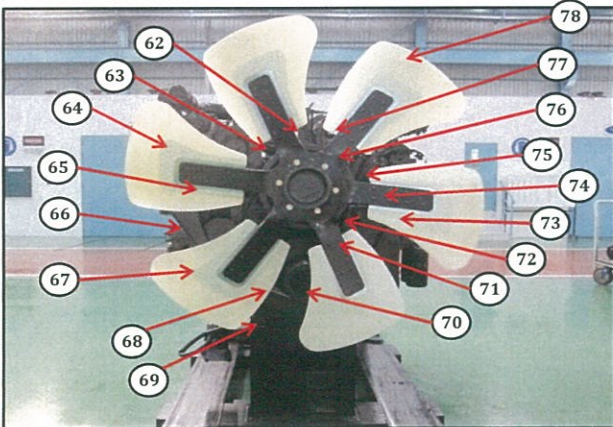
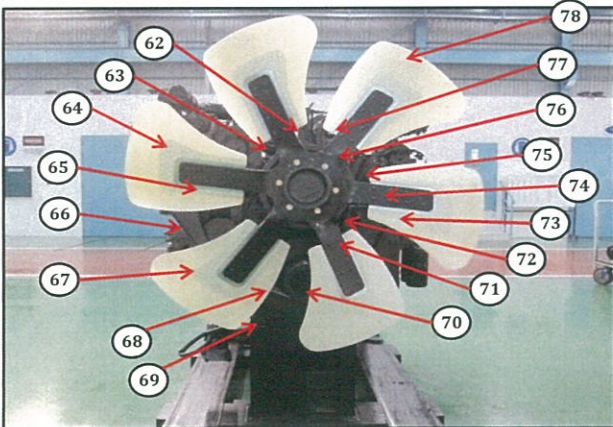
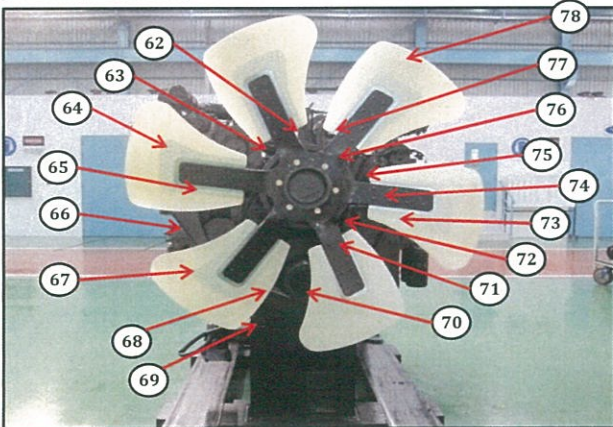
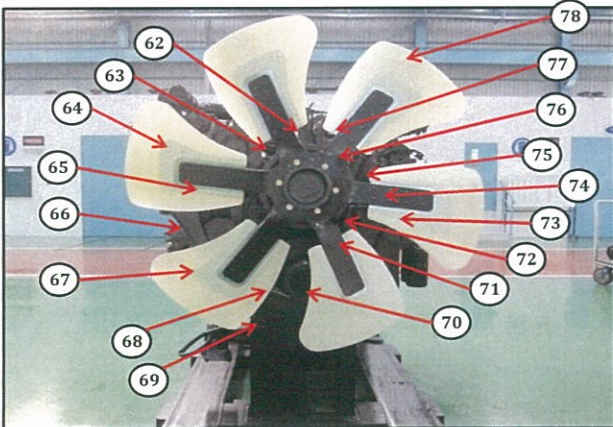
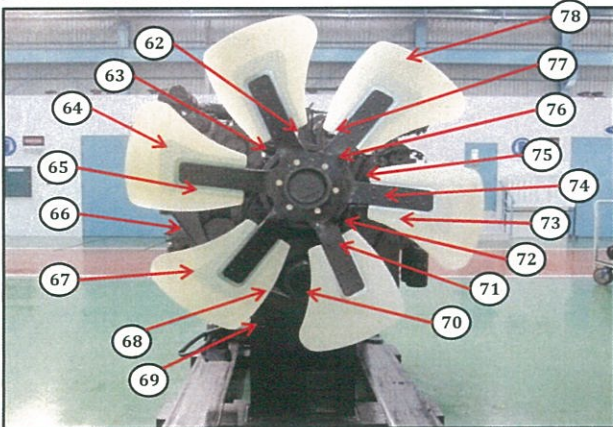
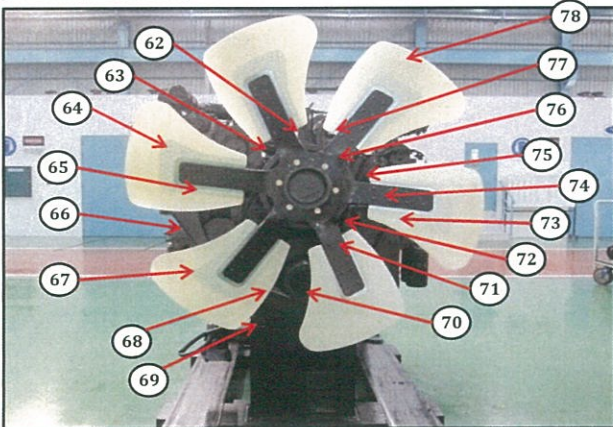
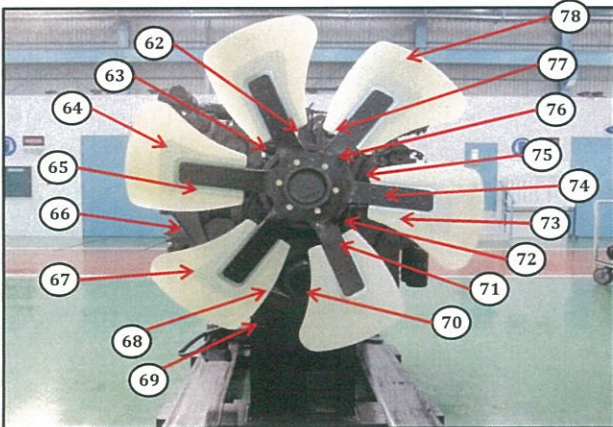
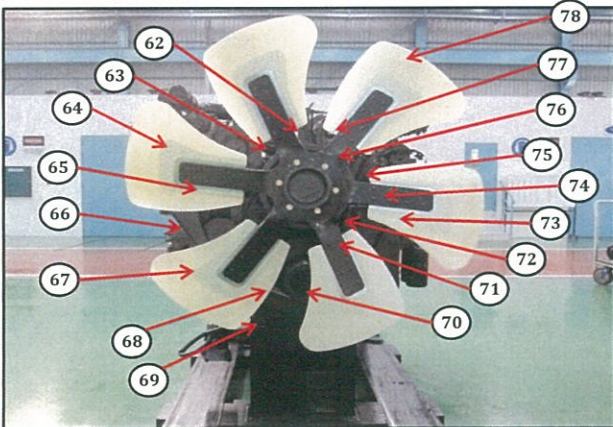
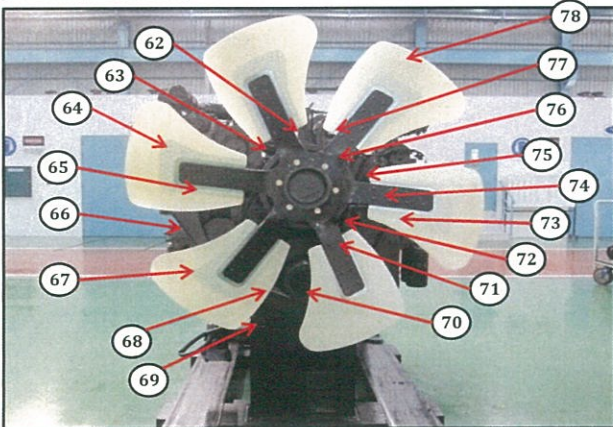
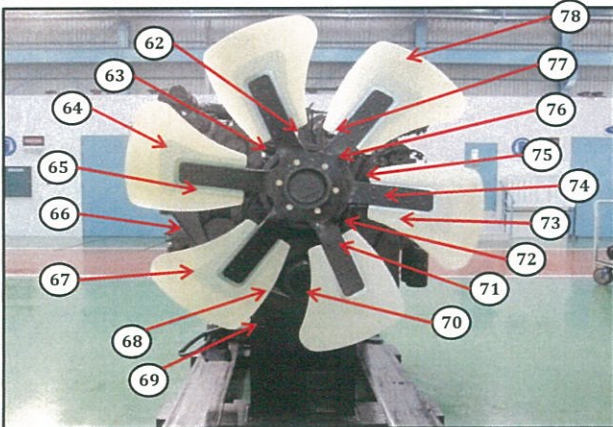
Notes :

RECEIVING INSPECTION SHEET ENGINE

MACHINE MODEL	HD785-7	ENGINE PART NO.		DOCUMENT NO.	RC/QR01/EG/01/01
ENGINE MODEL	SAA12V140E-3	WO ORDER NO.		PUBL. DATE	
ENGINE NO.		RECEIVING DATE		REVISION NO.	01
CUSTOMER NAME				PAGE	2 OF 2

BELOW ARE THE ITEMS RELATED TO DELIVERY INSPECTION AND WHILE YOU DOING THIS INSPECTION JOB, PLEASE REFERRING TO THE REMANUFACTURED COMPONENTS - STANDARD CONFIGURATION.

✓ : GOOD CONDITION X : BAD CONDITION ⊗ : NONE

SKETCH DRAWING OR PHOTOES	NO.	ITEM	INSPECTION CONDITION
Right Side View 	42	Air Connector RH	
	43	Exhaust Brake Assy RH	
Front Side View 	44	Heater Switch Assy With Cover RH	
	45	Air Intake Manifold RH	
Front Side View 	46	Oil Cooler Assy	
	47	Tube Cooling System Piping	
Front Side View 	48	Fuel Pressure Sensor With Limiter Assy RH	
	49	Common Rail Assy	
Front Side View 	50	Priming Pump Assy With Cover RH	
	51	Fuel Supply Pump Assy RH (P/N : 6219-71-1120)	
Front Side View 		S/N :	
	52	Oil Pan	
Front Side View 	53	Water Inlet Cover Plate	
	54	Water Pump Assy	
Front Side View 	55	Fuel Filter Assy	
	56	Cover Safety Guard Alternator	
Front Side View 	57	Tube, Breather With Hose RH	
	58	Oil Filter Assy	
Front Side View 	59	Alternator Assy (90A) (P/N : 600-825-9330)	
		S/N :	
Front Side View 	60	Tube Air Intake Connection RH	
	62	Plate Front Hanger	
Front Side View 	63	Fan Drive With Pulley Assy	
	64	Block Fuel Return With Sensor Assy RH & LH	
Front Side View 	65	V-Belt Alternator	
	66	Alternator Drive Pulley	
Front Side View 	67	Tension Pulley Assy	
	68	Vibration Damper Assy (P/N : 6215-31-8200)	
Front Side View 	69	Front Support	
	70	Crank Pulley Assy	
Front Side View 	71	Pointer	
	72	Yoke	
Front Side View 	73	V-Belt Fan Set	
	74	Blowbay Sensor Assy	
Front Side View 	75	Water Temperature Sensor	
	76	Thermostat Housing	
Front Side View 	77	Tube Water Outlet (2 pcs)	
	78	Cooling Fan	

Notes :

Inspected by,

Aproved by,



PT SAPTAINDRA SEJATI

ENGINE ASSY

DISASSEMBLY

CHECKSHEET

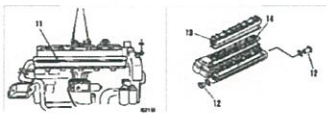
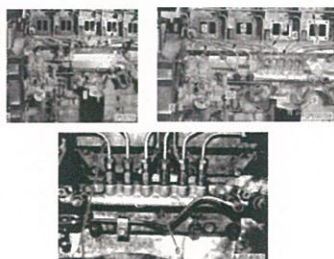
KOMATSU ENGINE 12V140E-3 SERIES

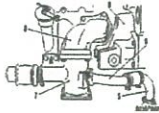
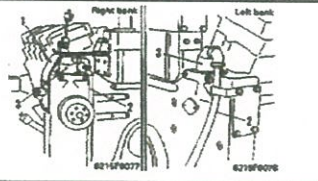
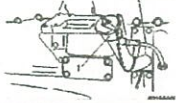
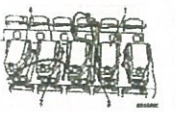
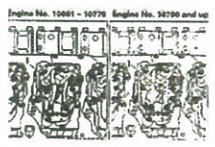
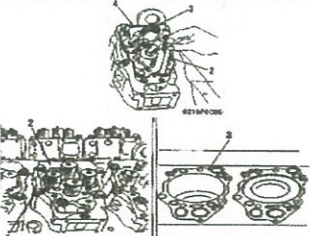
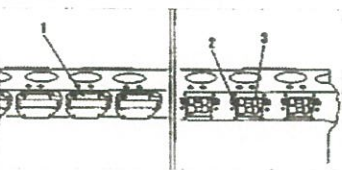
MACHINE MODEL

HD785-7

RO NO :

ENGINE SECTION

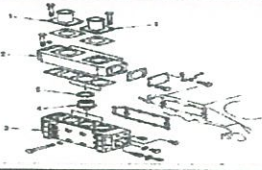
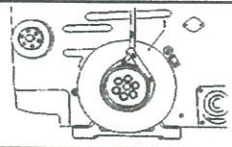
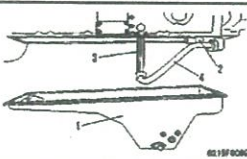
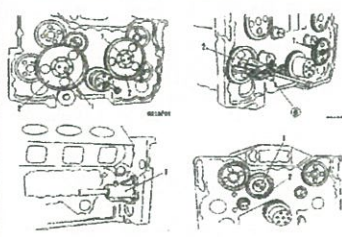
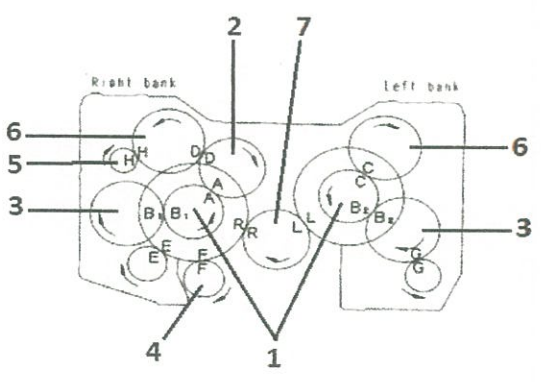
DISASSEMBLY CHECK SHEET KOMATSU ENGINE 12V140-3		WO NUMBER	UNIT CODE		
		ENGINE MODEL	PUBL. DATE		
		ENGINE SER. NO	REVISION NO.		
		MACHINE MODEL	DOCUMENT NO.		
PROCESS		START/date&time	DISASSEMBLY BY		
		FINISH/date&time			
NO.	PARTS TO REMOVE OR INSPECT	CONDITIONS TO BE INSPECTED	REPLACE SALVAGE REUSE	SKETCH DRAWING,OR PHOTOES	REMARKS
1	Engine ass'y	- Setting engine to engine stand ~ Prepare to stable engine stand that will prevent the engine from falling over, the set the engine assembly on the stand and secure it in position. - Drain engine oil from oil pan (the amount of oil differs according to the machine model) Rotate Crankshaft 720° for check Engine jammed or not.			Check Rotate Crankshaft YES: ☐ NO : ☐
2	Fuel Filter and Filter Head Ass'y, and Priming Pump, Corrosion Resistor	- Remove fuel tube, inspect for damage - Remove fuel filter, and filter head ass'y Inspect for threads and damage to fittings. - Remove priming pump - Remove corrosion resistor, tubes and mount. Inspect for damage. - Inspect the fuel filter housing for cracks, pulled threads and damage to fittings.			
3	Engine controller and controller cooler	- Remove connector, than remove engine controller - Remove fuel hoses - Remove controller cooler check for damage CONTROLLER RH,ESN: CONTROLLER LH,ESN:			
4	Starter Motor Alternator	- Please note if not fitted to engine. UPPER S/N: LOWER S/N: - Please note if not fitted to engine. ALTERNATOR S/N:	UPPER LOWER	CHECK BY SUB ASSY 3 	
5	Turbocharger	- Remove Turbocharger, mounting bolts & Adapters - Remove turbo oil line fittings. Check pipes and connections Turbocharger RH S/N: Turbocharger LH S/N:	LH RH	CHECK BY SUB ASSY 1 	
6	Exhaust Manifold	- Remove and inspect for cracks, pulled threads, snapped studs and missing bolts. Doing Colour Check inspection joint with QA Officer	FRONT RH CENTRE REAR FRONT LH CENTRE REAR		
7	After Cooler	- Check for pulled threads and missing bolts. - Remove aftercooler and check for damage, water leaks, dust entry etc. - Check aftercooler water tubes for cracks, corrosion and any other damage.	LH RH	CHECK BY SUB ASSY 2 	
8	Common Rail and Fuel Filter ass'y	- Remove all lines and clamps. Check for wear and damage. - Remove Common rail assembly and check for damage and cracks. - Remove and inspect the fuel filter housing and lines or cracks, pulled threads and damage to fittings.			

DISASSEMBLY CHECK SHEET KOMATSU ENGINE 12V140-3			WO NUMBER	UNIT CODE		
			ENGINE MODEL	PUBL. DATE		
			ENGINE SER. NO	REVISION NO.		
			MACHINE MODEL	DOCUMENT NO.		
NO.	PARTS TO REMOVE OR INSPECT	CONDITIONS TO BE INSPECTED	REUSE	REPLACE	SKETCH DRAWING, OR PHOTOES	REMARKS
9	Fuel Supply Pump ass'y	- Remove pump and mount, and inspect for damage, missing parts and wear. - Check fuel pump oil lines for damage and stripped threads. - Remove and check that fuel pump supply and return lines are in good condition. - Check the fuel pump drive couplings for signs of wear. <u>Fuel pump RH S/N:</u> <u>Fuel pump LH S/N:</u>			CHECK BY FIP  8) Using jig O, push out the fuel supply pump shaft to take out the fuel supply pump assembly. 	
10	Water Pump	- Check for evidence of leakage. - Check for signs of wear and damage.			 	
11	Breather	- Remove and inspect the breather and tube - The water pump is held by two mounting bolts at the front and two mounting bolts at the rear	LH RH			
12	Oil Cooler	- Check oil cooler housing for cracks and pulled threads. - Check oil cooler for dents and leaks.				
13	Oil Filler and Oil Level Gauge	Remove oil filler and oil level gauge, check for dents and cracks				
14	Rocker Covers and Rocker Housing	Remove Rocker Arm cover and check for cracks and damage to seal face. <u>Note :</u>			CHECK BY SUB ASSY 1  	
15	Rocker Arm	- Remove rocker arms and shafts and check for wear and other damage. - Check the contact area on the crossheads are from scuffing and wear. - Check the pushrods for bends and other damage.				
16	Cylinder Head	- Check cylinder head bolts for pin punch marks. If there are five marks, renew bolts. - Check the exhaust face condition and bolt holes for pulled threads. - Check the head surface for water marks or damage. - Doing Colour Check inspection joint with QA Officer				
17	Cam Followers	- Inspect cam followers for damage to the contact surface of the rollers. - Check condition of covers. - The mounting bolts of the cam follower is a special bolt, so keep it in a safe place and be careful not to lose it				

DISASSEMBLY CHECK SHEET

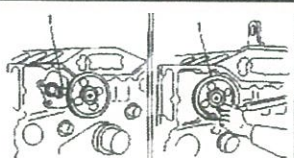
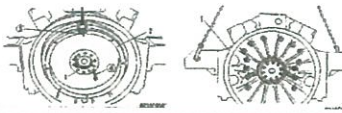
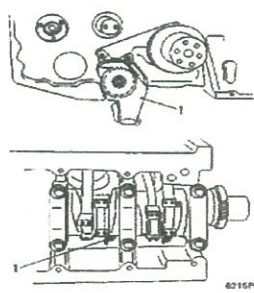
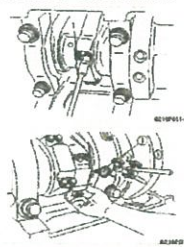
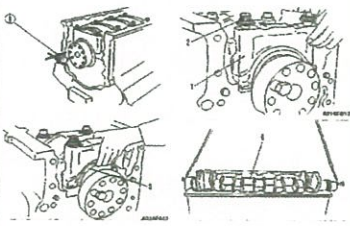
KOMATSU ENGINE 12V140-3

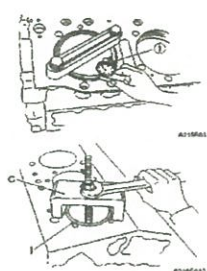
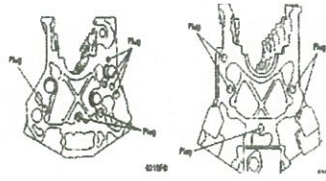
WO NUMBER		UNIT CODE	
ENGINE MODEL	SAA12V140E-3	PUBL. DATE	
ENGINE SER. NO		REVISION NO.	06
MACHINE MODEL		DOCUMENT NO.	

NO.	PARTS TO REMOVE OR INSPECT	CONDITIONS TO BE INSPECTED	REUSE	REPLACE	SKETCH DRAWING, OR PHOTOS	REMARKS
18	Thermostat housing Thermostat case	- Disconnect thermostat case and thermostat housing and check condition for cracked and damage. - Check surface contact thermostat				
19	Vibration Damper and Pulleys	- Check vibration damper for damage. - Check pulleys for damage and wear. - Check belt tensioner for wear. - Check front trunnion mount.				
20	Oil Pan	- Remove pan and check for dents and cracks. - Check oil level sensor - Check drain plug for pulled threads - Check oil pick up pipe for cracks in the welds - Check screen for damage.				
21	Front Support and Timing Cover	- Remove and inspect front cover for damage - Check surface condition of seal running area on front shaft.				
22	Timing Gears	- Inspect and record condition of individual gears. Check for chipped teeth, cracks, excessive backlash and end float. - Check shafts and bearings for wear. - Measure thickness of thrust plates for wear. - Check alternator drive. - Check fuel pump drive. - Check accessory drive. - Check water pump drive.			 	
Unit : mm						
Backlash		Measured	End Play		Measured	
R-R	0.141-0.425		1	0.07 - 0.18	RH	
A-A	0.030-0.390		2	0.07 - 0.18	LH	
D-D	0.129-0.391		3	0.07 - 0.20	RH	
H-H	0.085-0.492		4	0.03 - 0.088	LH	
B1-B1	0.051-0.469		5	0.1 - 0.4		
E-E	0.052-0.481		6	0.14 - 0.32		
F-F	0.137-0.421		7	0.10 - 0.25	RH	
L-L	0.141-0.425				LH	
C-C	0.129-0.391					
B2-B2	0.051-0.469					

DISASSEMBLY CHECK SHEET KOMATSU ENGINE 12V140-3

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MACHINE MODEL		DOCUMENT NO.	

NO.	PARTS TO REMOVE OR INSPECT	CONDITIONS TO BE INSPECTED	REUSE	REPLACE	SKETCH DRAWING, OR PHOTOS	REMARKS														
23	Camshaft	- Check camshaft journals for wear. - Check end float of camshaft. - Check cam surfaces for wear and other damage - Check camshaft for bends.																		
24	Flywheel and Flywheel Housing	- Check bolts and bolt holes. - Check pilot bearing hole. - Inspect ring gear for chipped and worn teeth. - Check housing for cracks, check bolt holes.																		
25	Oil Pump Ass'y Piston Cooling Nozzles	- Check pump O-rings at inlet and outlet ports. - Check pump for wear and damage. - grooves, skirts and clogde <table border="1"> <tr> <td>RH</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr> <tr> <td>LH</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr> </table> Notes : ☐ Good ☐ No Good	RH	1	2	3	4	5	6	LH	1	2	3	4	5	6				
RH	1	2	3	4	5	6														
LH	1	2	3	4	5	6														
26	Connecting Rods	- Check con rod for cracks and damage. - Check big end bore for scratches or distortion. - Check for damage on mating face of cap. <table border="1"> <tr> <td>RH/LH</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr> <tr> <td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr> </table> - Standard side clearance (End Play) 0.300 ~ 0.454 mm	RH/LH	1	2	3	4	5	6										CHECK BY MACHINING 	
RH/LH	1	2	3	4	5	6														
27	Pistons	Measure wear on piston skirt, ring grooves and pin bore <table border="1"> <tr> <td>RH</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr> <tr> <td>LH</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr> </table> Notes : ☐ Good, ☐ No Good	RH	1	2	3	4	5	6	LH	1	2	3	4	5	6				
RH	1	2	3	4	5	6														
LH	1	2	3	4	5	6														
28	Crankshaft	- Inspect main and pin journals for scratches, wear, cracks and discoloration. - Check that oil holes are not blocked. - Check fillet area for scratches. - Determine wear by measuring $\varnothing$ of journals. - Check seal surface condition. - Check gear of crankshaft from pitting, chipping and highly scratch. Notes: ☐ Good, ☐ No Good Gear Broken : ☐ Gear no broken: ☐ Marking S/N engine on rear Crankshaft (marking on free area or no contact area with flywheel) Crankshaft S/N:																		

DISASSEMBLY CHECK SHEET KOMATSU ENGINE 12V140-3			WO NUMBER	UNIT CODE																							
			ENGINE MODEL	PUBL. DATE																							
			ENGINE SER. NO	REVISION NO.																							
			MACHINE MODEL	DOCUMENT NO.																							
NO.	PARTS TO REMOVE OR INSPECT	CONDITIONS TO BE INSPECTED	REPLACE SALVAGE REUSE	SKETCH DRAWING, OR PHOTOS	REMARKS																						
29	Measuring Crankshaft	- Measure Endplay of Crankshaft - Measure Face & Radial out of Crankshaft at front and rear surface Crankshaft before remove Crankshaft assy	Maximum 0.05mm (Reference: STD Alignment main bearing bore)	1. CRANKSHAFT ENDPLAY ENDPLAY: mm 2. FACE RUN OUT FRONT FR/OUT: mm REAR FR/OUT: mm 3. RADIAL RUN OUT FRONT RADIAL: mm REAR RADIAL: mm																							
30	Main Cap stamping	- Check cap bolts for number of punch marks. - Check cap fit to block. Punch S/N engine at the front face of all main bearing cap match with S/N engine block		 																							
31	Main cap Thrust metal seat	- Check contact main cap with cylinder block at Thrust metal seat. Contact must flat (no different with cap and cyl block). - Thrust metal seat no. 7 main cap for engine 12V140 and no. 6 cap for engine 170-3/-5 series.		 																							
32	Cylinder Liners	- Check for pitting and cracks on the outer circumference of the Cylinder liner - Inspect the inner surface for scoring and corrosion. - Check for pitting damage on seal grooves. - Measure inside Ø, roundness and cylindricity of Cylinder liner <table border="1"> <tr> <td></td> <td>1</td> <td>2</td> <td>3</td> <td>4</td> <td>5</td> <td>6</td> </tr> <tr> <td>RH</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>LH</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> </table> Notes : ☒ Good ☐ No Good		1	2	3	4	5	6	RH							LH										
	1	2	3	4	5	6																					
RH																											
LH																											
33	Cylinder Block and Check interference cap to block one by one	- Check for cracks in the block, especially in the bolt hole part of the cylinder head. - Inspect the lower liner seal area. - Check the blind plug surface for corrosion. - Check all threads. - Check mounting surface of cylinder head for distortion and corrosion. - Check for clogging of oil and water holes. - Remove all Camshaft bushing before Colour check * Doing Colour Check inspection joint with QA Officer * After finish colour check all Main cap must be Torque 58kgm+90° by Disassy mechanic * Move the Cyl. Block with using hanger, than torque hanger Mounting Bolt. <table border="1"> <tr> <td>STD. Torque</td> <td>7 Kg.m.</td> </tr> <tr> <td>Actual</td> <td></td> </tr> </table>	STD. Torque	7 Kg.m.	Actual			CHECK BY MACHINING   INSPECTION CHECK INTERFERENCE CAP TO BLOCK ONE BY ONE <table border="1"> <tr> <td>CAP</td> <td>1</td> <td>2</td> <td>3</td> <td>4</td> <td>5</td> <td>6</td> </tr> <tr> <td>CHECK</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> </table> <table border="1"> <tr> <td>OK</td> <td>☒</td> </tr> <tr> <td>NOT OK</td> <td>☐</td> </tr> </table>	CAP	1	2	3	4	5	6	CHECK							OK	☒	NOT OK	☐	
STD. Torque	7 Kg.m.																										
Actual																											
CAP	1	2	3	4	5	6																					
CHECK																											
OK	☒																										
NOT OK	☐																										

Notes :

INSPECTION BY

APPROVAL BY

MECHANIC

GL/SPV PRODUCTION



PT SAPTAINDRA SEJATI

**CRANKSHAFT DISASSEMBLY
INSPECTION AND MEASUREMENT
CHECKSHEET**

(BEFORE MACHINING)

KOMATSU ENGINE 12V140E-3 SERIES

MACHINE MODEL

HD785-7

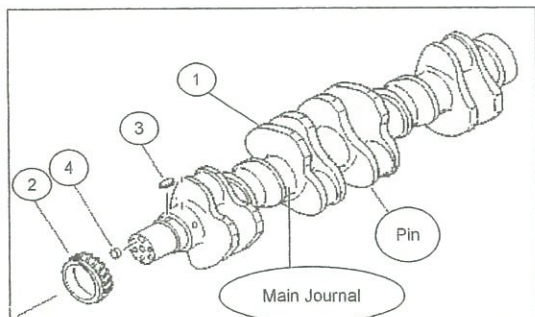
RO NO :

ENGINE SECTION

CRANK SHAFT DISASSEMBLY INSPECTION AND MEASUREMENT CHECKSHEET

WO NUMBER		PART NAME	CRANKSHAFT ASSY	DOCUMENT NO.	QR08-SAL-011-01
ENGINE MODEL	12V140-1/-3	PART NUMBER	6215-31-1101	PUBL. DATE	01/04/2009
ENGINE S/N		START DATE / TIME		REVISION NO.	01
MACHINE MODEL		FINISH DATE / TIME		PAGE	1 of 1

SKETCH DRAWING



Physical Data

No.	Part Name	Qty	Good	No Good
1	Crankshaft	1		
2	Gear	1		
3	Key	1		
4	Dowel Pin	1		

Standard size

Unit : mm

SIZE	MAIN JOURNAL		PIN JOURNAL	
	STD SIZE	REPAIR LIMIT	STD SIZE	REPAIR LIMIT
STANDARD	148.000 ~ 147.980	147.980	100.000 ~ 99.980	99.980
Under Size 0.25	147.750 ~ 147.730	147.730	99.750 ~ 99.730	99.730
Under Size 0.50	147.500 ~ 147.480	147.480	99.500 ~ 99.480	99.480
ROUNDNESS	0 ~ 0.010		0 ~ 0.010	
FILLET RADIUS	3.0 ~ 3.5	-	4.0 ~ 4.5	-

MANHOURS UTILIZATION

PROCESS	START	FINISH	MAN HOURS
Washing / Cleaning			
Measuring Process			
TOTAL			

VISUAL CHECK

OK

Not OK

Crack, Scratch, Wear Has been checked

FILLET RADIUS

Main Journal

Pin Journal

INSPECTION AND MEASUREMENT BY QA & MECHANIC DIS-ASSEMBLY

CRITICAL INSPECTION		RANDOM INSPECTION		ROUNDNESS	CRITICAL INSPECTION FILLET RADIUS		RANDOM INSPECTION		REMARK
		CHECK POINT					CHECK POINT		
		X - X	Y - Y				ACTUAL MEASUREMENT		
MAIN JOURNAL	1				MAIN JOURNAL	1			
	2					2			
	3					3			
	4					4			
	5					5			
	6					6			
	7					7			
PIN JOURNAL	1				PIN JOURNAL	1			
	2					2			
	3					3			
	4					4			
	5					5			
	6					6			

Main Journal & Pin Journal

Actual : STD or US

Crank Shaft Condition : Good or No Good

Inspection result :

MEASURING BY DIS ASSY

MEASURING DATE :

MECHANIC NAME :

SIGN :

RANDOM INSPECTION BY QA

MEASURING DATE :

INSPECTOR NAME :

SIGN :

DISASSEMBLY CHECK SHEET KOMATSU ENGINE 12V140-3

WO NUMBER		UNIT CODE	
ENGINE MODEL	SAA12V140E-3	PUBL. DATE	
ENGINE SER. NO		REVISION NO.	06
MACHINE MODEL		DOCUMENT NO.	

ATTACHED

OUTSIDE DIAMETER OF CRANKSHAFT PIN JOURNAL AND MAIN JOURNAL

6D170 - 3 Series

Unit : mm CHECK BY MACHINING

Size	Main Journal		Pin Journal	
	Standard Size	Reman Limit	Standard Size	Reman Limit
Standard	140,000	139,910	108,000	107,910
0.25 US	139,750	139,660	107,750	107,660
0.50 US	139,500	139,410	107,500	107,410
0.75 US	139,250	139,160	107,250	107,160
1.00 US	139,000	139,910	107,000	106,910
Out of Roundness	0 - 0.010	0,010	0 - 0.010	0,010
Tolerance :	- 0.025			

6D140 - 3 Series

Unit : mm

Size	Main Journal		Pin Journal	
	Standard Size	Reman Limit	Standard Size	Reman Limit
Standard	120,000	119,830	90,000	89,880
0.25 US	119,750	119,630	89,750	89,630
0.50 US	119,500	119,830	89,500	89,380
0.75 US	119,250	119,130	89,250	89,130
1.00 US	119,000	118,880	89,000	88,880
Out of Roundness	0 - 0.010	0,020	0 - 0.010	0,020
Tolerance :	- 0.020			

12V140 - 3 Series

Unit : mm

Size	Main Journal		Pin Journal	
	Standard Size	Reman Limit	Standard Size	Reman Limit
Standard	148,000 ~ 147,980		100,000 ~ 99,980	99,880 and less
0.25 US	147,750 ~ 147,730		99,750 ~ 99,730	99,630 and less
0.50 US	147,500 ~ 147,480		99,500 ~ 99,480	99,380 and less
0.75 US	147,250 ~ 147,230		99,250 ~ 99,230	99,130 and less
1.00 US	147,000 ~ 146,980		99,000 ~ 98,980	98,880 and less
Out of Roundness	0.010 and less	0.011 and more	0.010 and less	0,011 and more
Tolerance :	- 0.020			

OUTSIDE DIAMETER OF CAMSHAFT JOURNAL AND CAM HEIGHT

12V140 - 3 Series

Cam Height	Std Size	Tolerance	Repair limit
Intake	55,48	± 0.1	55,08
Exhaust	55,75	± 0.1	55,35
OD Chamshaft Journal	Std Size	Tolerance	Repair limit
STD	65	-0.016 -0.036	

6D140-3 Series

Unit : mm

Cam Height	Std Size	Tolerance	Repair limit
Intake	55,48	± 0.1	55,08
Exhaust	55,75	± 0.1	55,35
OD Chamshaft Journal	Std Size	Tolerance	Repair limit
STD	65	-0.016 -0.036	-

6D170 - 3 Series

Unit : mm

Cam Height	Std Size	Tolerance	Repair limit
Intake	89,8	+0.1081 -0.0919	89,20
Exhaust	90,1	+0.1688 -0.0312	88,80
OD Chamshaft Journal	Std Size	Tolerance	Repair limit
STD	105	-0.042 -0.072	104,88



PT SAPTAINDRA SEJATI

MEASURING CYLINDER BLOCK BEFORE SALVAGING CHECKSHEET

KOMATSU ENGINE 12V140E-3 SERIES

MACHINE MODEL

HD785-7

RO NO :

ENGINE SECTION

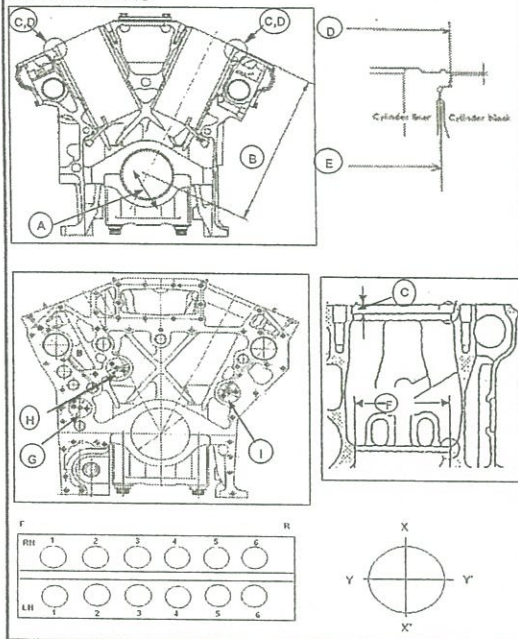


PT SAPTAINDRA SEJATI
NAROGONG REBUILD CENTER

MEASURING CYLINDER BLOCK BEFORE SALVAGING CHECKSHEET

JOB NUMBER		PART NAME	CYLINDER BLOCK	DOCUMENT NO.	RC/QA/EG/018/R05
ENGINE MODEL	SA12V140 - 1/3	PART NUMBER		PUBL. DATE	15-05-2018
ENGINE SER.NO.		START DATE / TIME		REVISION NO.	05
MACHINE MODEL		FINISH DATE / TIME		PAGE	

SKETCH DRAWING



MECHANIC NAME	Cyl. Block Inspection	
	Cyl. Block assembly	
Standard Size		Unit : mm
DESCRIPTION	STANDARD SIZE	
A	MAIN BEARING BORE	157.000 ~ 157.025
B	CYLINDER BLOCK HEIGHT	471.460 ~ 471.540
	USE OVER SIZE GASKET	471.062 ~ 471.450
C	COUNTER BORE DEPTH	11.00 ~ 11.05
D	INSIDE DIAMETER (AT FLANGE SEAT)	170.200 ~ 170.263
E	INSIDE DIAMETER (AT COUNTERBORE)	161.170 ~ 161.230
F	INSIDE DIAMETER (AT O-RING LAND)	VISUAL CHECK
G,H,I	TIMING GEAR HOLE	60.000 ~ 60.030

TIGHTENING TORQUE OF MAIN BEARING CAP	1 ST step = 27.5 ~ 30.5 Kgm
	2 ND step = 57.0 ~ 59.0 Kgm
	3 RD step = 90° ~ 120°

VISUAL CHECK	OK	Not OK
CRACK HAS BEEN CHECKED		

MANHOURS UTILIZATION		
START	FINISH	MAN HOURS
Measuring Process		
TOTAL		

MAIN BEARING BORE

Main Bore No.		X - X'	Y - Y'	Z - Z'
A	1 A			
	B			
	2 A			
	B			
	3 A			
	B			
	4 A			
	B			
	5 A			
	B			
	6 A			
	B			
	7 A			
	B			

COUNTER BORE DEPTH

BEFORE SALVAGING												
C	R.H Bank Front	Cyl	Mech				L.H Bank Front	Cyl	Mech			
			X	X'	Y	Y'			X	Y	Y'	Y''
		1						1				
		2						2				
		3						3				
		4						4				
		5						5				
		6						6				

JOB NUMBER		PART NAME	CYLINDER BLOCK	DOCUMENT NO.	RC/QA/EG/018/R05
ENGINE MODEL	SA12V140 - 1/3	PART NUMBER		PUBL. DATE	15-05-2018
ENGINE SER.NO.		START DATE / TIME		REVISION NO.	05
MACHINE MODEL		FINISH DATE / TIME		PAGE	

C	COUNTER BORE DEPTH	11.00 ~ 11.05
D	INSIDE DIAMETER (AT FLANGE SEAT)	170.200 ~ 170.263
E	INSIDE DIAMETER (AT COUNTERBORE)	161.170 ~ 161.230

INSIDE DIAMETER(AT FLANGE SEAT)

BEFORE SALVAGING										
D	R.H Bank	Cyl	X - X'			Cyl	Y - Y'			
			CHECK				CHECK			
			MEC	QA/GL	SPV		MEC	QA/GL	SPV	
		1				1				
		2				2				
		3				3				
		4				4				
	5				5					
	6				6					
	L.H Bank	Cyl	X - X'			Cyl	Y - Y'			
CHECK			CHECK							
			MEC	QA/GL	SPV			MEC	QA/GL	SPV
1							1			
2							2			
3							3			
4							4			
5					5					
6					6					

TIMING GEAR HOLE(VISUAL CHECK)

TIMING GEAR HOLES						
G,H,I	X - X'			Y - Y'		
	Mac	QA/GL	Spv	Mac	QA/GL	Spv

INSIDE DIAMETER(AT COUNTERBORE)

BEFORE SALVAGING									
E	RH	Cyl	X - X'			Cyl	Y - Y'		
			CHECK				CHECK		
		MEC	QA/GL	SPV	MEC	QA/GL	SPV		
		1				1			
		2				2			
		3				3			
		4				4			
	5				5				
	6				6				
		LH	Cyl	X - X'			Cyl	Y - Y'	
CHECK				CHECK					
MEC			QA/GL	SPV	MEC	QA/GL	SPV		
1						1			
2						2			
3						3			
4						4			
5					5				
6					6				

JOB NUMBER		PART NAME	CYLINDER BLOCK	DOCUMENT NO.	RC/QA/EG/018/R05
ENGINE MODEL	SA12V140 - 1/3	PART NUMBER		PUBL. DATE	15-05-2018
ENGINE SER.NO.		START.DATE / TIME		REVISION NO.	05
MACHINE MODEL		FINISH DATE / TIME		PAGE	

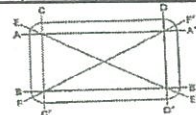
F	INSIDE DIAMETER (AT O-RING LAND)	VISUAL CHECK
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INSIDE DIAMETER (AT O-RING LAND)					
F	RH	Cyl	VISUAL CHECK		
			CHECK		
		MEC	QA/GL	SPV	
		1			
		2			
		3			
		4			
		5			
		6			
		LH	LH	Cyl	VISUAL CHECK
CHECK					
MEC	QA/GL			SPV	
1					
2					
3					
4					
5					
6					

Judgement of Distortion (Flatness) : Measure thickness between Block & Straight edge.

Flatness	Remedy
within 0.05mm	Correct with surface grinding

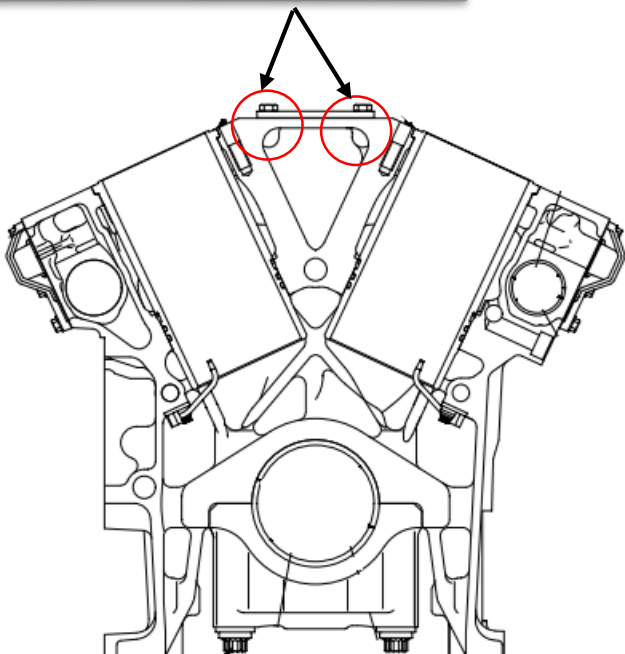
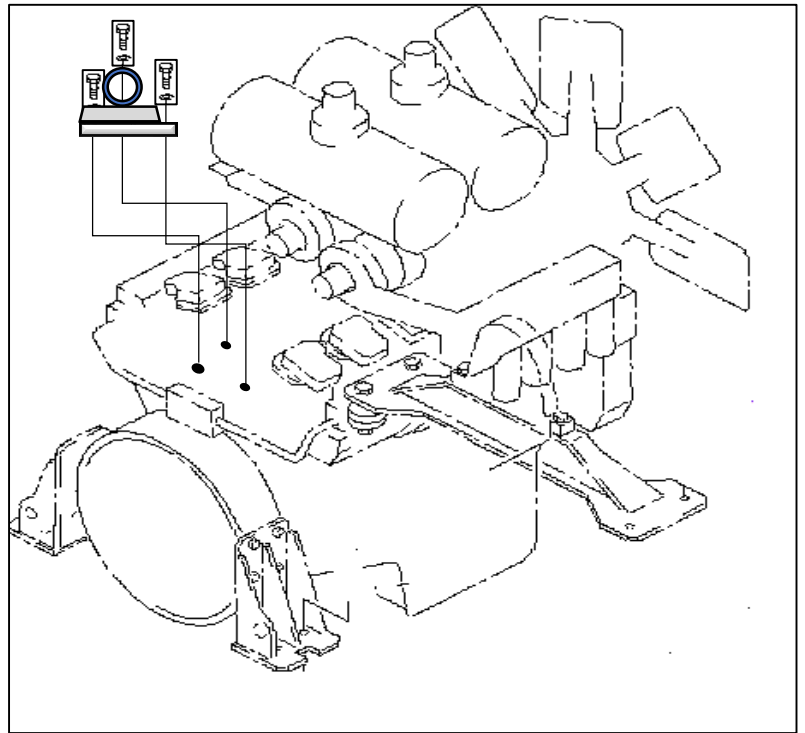
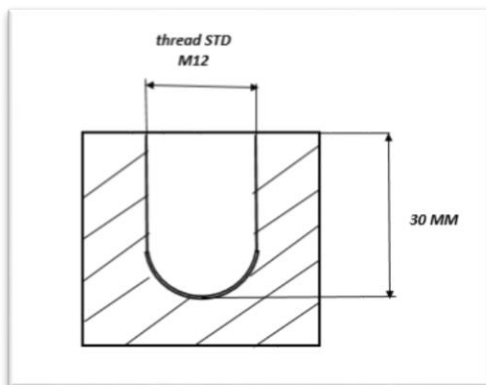
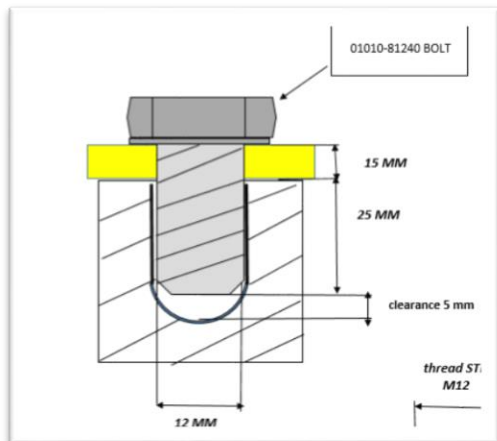
Measurement tool : Straight edge, clearance gauge

[illegible]

INSPECTION RESULT :		
MEASURED	INSPECTION & APPROVAL BY :	
_____ MECHANIC	_____ QUALITY ASSURANCE (QA)	_____ GL/SPV PRODUCTION

WARNING

PROSES LIFTING ENGINE OR
CYLINDER BLOCK 12V140 SERIES





PT SAPTAINDRA SEJATI

CAMSHAFT

POLISHING PROCESS AND MEASUREMENT

CHECKSHEET

KOMATSU ENGINE 12V140E-3 SERIES

MACHINE MODEL

HD785-7

RO NO :

ENGINE SECTION

CAMSHAFT PROCESS AND MEASUREMENT CHECK SHEET

JOB NUMBER		PART NAME	CAMSHAFT RH	DOCUMENT NO.	
ENGINE MODEL	12V140-3	PART NUMBER		PUBL. DATE	RC/QR03/EG/07/01
ENGINE S/N		START DATE/TIME		REVISION NO.	03
MACHINE MODEL		FINISH DATE/TIME		PAGE	1 of 1

MACHINIST NAME	
----------------	--

MAN HOURS UTILIZATION			
	START	FINISH	MAN HOURS
Machining Process			
Measuring Process			
TOTAL			



Right bank Camshaft



Left bank Camshaft

Unit : mm

STANDARD		REPAIR LIMIT	No		CAMSHAFT JOURNAL			CAM LUBE INTAKE			CAM LUBE EXHAUST		
					MEC	QA	SPV	MEC	QA	SPV	MEC	QA	SPV
12V140	Intake = 55.58 ~ 55.38	55,08	1L	X									
	Exhaust = 55.65 ~ 55.85	55,35		Y									
	Cam Journal = 64.984 ~ 64.964		2L	X									
				Y									
6D140	Intake = 55.58 ~ 55.38	55,08	3L	X									
	Exhaust = 55.85 ~ 55.65	55,35		Y									
	Cam Journal = 64.984 ~ 64.964		4L	X									
				Y									
6D170-3	Intake = 91.240 ~ 91.040	89,20	5L	X									
	Exhaust = 90.268 ~ 90.068	88,80		Y									
	Cam Journal = 104.958 ~ 104.928	104,88	6L	X									
				Y									
6D170-2	Intake = 62.414 ~ 62.214	61,37	7L	X									
	Exhaust = 60.678 ~ 60.478	59,64		Y									
	Cam Journal = 71.92 ~ 71.89	71,86	1R	X									
				Y									
DESCRIPTION	OK	Not OK	2R	X									
				Y									
CRACK HAS BEEN CHECK			3R	X									
				Y									
Right bank Camshaft			4R	X									
				Y									
Left bank Camshaft			5R	X									
				Y									
			6R	X									
				Y									
			7R	X									
				Y									

INSPECTION RESULT :

INSPECTION BY :

INSPECTION & APPROVED BY :

MECHANIC(MEC)

QUALITY ASSURANCE (QA)

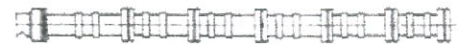
GL/SPV PRODUCTION

CAMSHAFT PROCESS AND MEASUREMENT CHECK SHEET

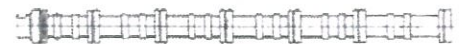
JOB NUMBER		PART NAME	CAMSHAFT LH	DOCUMENT NO.	
ENGINE MODEL	12V140-3	PART NUMBER		PUBL. DATE	RC/QR03/EG/07/01
ENGINE S/N		START DATE/TIME		REVISION NO.	03
MACHINE MODEL		FINISH DATE/TIME		PAGE	1 of 1

MACHINIST NAME	
----------------	--

MAN HOURS UTILIZATION			
	START	FINISH	MAN HOURS
Machining Process			
Measuring Process			
TOTAL			



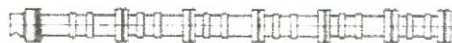
Right bank Camshaft



Left bank Camshaft

Unit : mm

STANDARD		REPAIR LIMIT	No		CAMSHAFT JOURNAL			CAM LUBE INTAKE			CAM LUBE EXHAUST		
					MEC	QA	SPV	MEC	QA	SPV	MEC	QA	SPV
12V140	Intake = 55.58 ~ 55.38	55,08	1L	X									
	Exhaust = 55.65 ~ 55.85	55,35		Y									
	Cam Journal = 64.984 ~ 64.964		2L	X									
				Y									
6D140	Intake = 55.58 ~ 55.38	55,08	3L	X									
	Exhaust = 55.85 ~ 55.65	55,35		Y									
	Cam Journal = 64.984 ~ 64.964		4L	X									
				Y									
6D170-3	Intake = 91.240 ~ 91.040	89,20	5L	X									
	Exhaust = 90.268 ~ 90.068	88,80		Y									
	Cam Journal = 104.958 ~ 104.928	104,88	6L	X									
				Y									
6D170-2	Intake = 62.414 ~ 62.214	61,37	7L	X									
	Exhaust = 60.678 ~ 60.478	59,64		Y									
	Cam Journal = 71.92 ~ 71.89	71,86	1R	X									
				Y									
DESCRIPTION	OK	Not OK	2R	X									
				Y									
CRACK HAS BEEN CHECK			3R	X									
				Y									
			4R	X									
				Y									
			5R	X									
				Y									
			6R	X									
				Y									
			7R	X									
				Y									



Right bank Camshaft



Left bank Camshaft

INSPECTION RESULT :

INSPECTION BY :

INSPECTION & APPROVED BY :

MECHANIC(MEC)

QUALITY ASSURANCE (QA)

GL/SPV PRODUCTION



PT SAPTAINDRA SEJATI

CONNECTING ROD

MEASUREMENT INSPECTION

CHECKSHEET

KOMATSU ENGINE 12V140E-3 SERIES

MACHINE MODEL

HD785-7

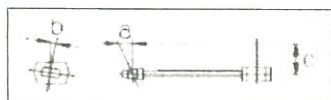
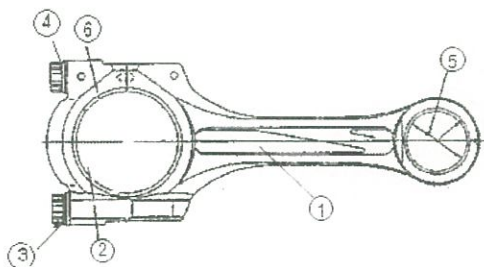
RO NO :

ENGINE SECTION

CONNECTING ROD SALVAGING PROCESS AND INSPECTION CHECKSHEET

WO NUMBER		PART NAME	CONNECTING ROD	EKS UNIT CODE	
ENGINE MODEL	SAA12V140E-3	PART NUMBER	6219-31-3100/3200	PUBL. DATE	24/08/2010
ENGINE S/N		START DATE / TIME		REVISION NO.	00_1 of 2
MACHINE MODEL		FINISH DATE / TIME		DOCUMENT NO.	

SKETCH DRAWING



Physical Data

MECHANIC NAME				
Standard size				
Unit : mm				
ENGINE MODEL	TIGHTENING BOLT STEP PROCESS WITHOUT BEARING (METAL) KGM + ANGLE / °	DIAMETER SMALL END WITHOUT BEARING	DIAMETER SMALL END WITH BEARING	REPAIR LIMIT
6D140 - 3	12.5 ~ 13.5 + 90 ° ~ 120 °	57.400 ~ 57.430	52.030 ~ 52.049	52,09
6D170 - 3	19.0 ~ 21.0 + 90 ° ~ 105 °	75.000 ~ 75.030	68.010 ~ 68.049	-
12V140-3	7.0 ~ 8.0 + 90 ° ~ 120 °	57.400 ~ 57.430	52.030 ~ 52.049	52,09
ENGINE MODEL	DIAMETER BIG END WITHOUT BEARING	BEND (a)	TWIST (b)	DISTANCE BETWEEN HOLE
6D140 - 3	94.992 ~ 95.042	RL 0.35	RL 0.25	264
6D170 - 3	115.000 ~ 115.026	RL 0.10	RL 0.25	305
12V140-3	105.996 ~ 106.026	RL 0.15	RL 0.30	340

No.	Part Name	Qty	New	Reuse	VISUAL CHECK		OK	NOT
1	Connecting rod				CRACK HAS BEEN CHECKED			
2	Connecting rod cap							
3	Connecting rod bolt							
4	Washer				MANHOURS UTILIZATION			
5	Crank pin Metal				Measuring Process	START	FINISH	MAN HOURS
					TOTAL			

BIG END BEFORE SALVAGING PROCESS (NEW BEARING)

CON-ROD NO.	TIGHTENING BOLTS	CHECK POINT	Ø BIG END WITHOUT BEARING					
			X - X'			Y - Y'		
			MAC	QA/GL	SPV	MAC	QA/GL	SPV
1R		A						
2R		A						
3R		A						
4R		A						
5R		A						
6R		A						
1L		A						
2L		A						
3L		A						
4L		A						
5L		A						
6L		A						

CONNECTING ROD SALVAGING PROCESS AND INSPECTION CHECKSHEET

WO NUMBER		PART NAME	CONNECTING ROD	EKS UNIT CODE	
ENGINE MODEL	SAA12V140E-3	PART NUMBER	6219-31-3100/3200	PUBL DATE	24/08/2010
ENGINE S/N		START DATE / TIME		REVISION NO.	00 2 of 2
MACHINE MODEL		FINISH DATE / TIME		DOCUMENT NO.	

SMALL END BEFORE SALVAGING PROCESS (NEW BEARING)

CON-ROD NO.	CHECK POINT	Ø SMALL END WITH BEARING					
		X - X'			Y - Y'		
		MAC	QA/GL	SPV	MAC	QA/GL	SPV
1R	A						
2R	A						
3R	A						
4R	A						
5R	A						
6R	A						
1L	A						
2L	A						
3L	A						
4L	A						
5L	A						
6L	A						

CON ROD BENDING

CON-ROD NO.	BEND		TWIST	
	Indicator Reading	Actual Bending	Indicator Reading	Actual Twist
1R				
2R				
3R				
4R				
5R				
6R				
1L				
2L				
3L				
4L				
5L				
6L				

BEND AND TWIST

Measure with formula :

$$\frac{\text{Distance between hole}}{\text{Width seat metal}} \times 1 \text{ Reading}$$

INSPECTION RESULT :

INSPECTION BY :

INSPECTION & APPROVAL BY :

MECHANIC (MECH)

QUALITY ASSURANCE (QA)

GL/SPV PRODUCTION



PT SAPTAINDRA SEJATI

PISTON PIN MEASURING & POLISHING PROCESS CHECKSHEET

KOMATSU ENGINE 12V140E-3 SERIES

MACHINE MODEL

HD785-7

RO NO :

ENGINE SECTION



PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

PISTON PIN MEASURING AND POLISHING CHECK SHEET

WO NUMBER		PART NAME	PISTON PIN	EKS UNIT CODE	
ENGINE MODEL	12V140-1,-3	PART NUMBER	6211-31-2410	PUBL. DATE	25/08/2009
ENGINE S/N		START DATE / TIME		REVISION NO.	00
MACHINE MODEL		FINISH DATE / TIME		DOCUMENT NO.	

MECHANIC NAME	
---------------	--

MAN HOURS UTILIZATION			
PROCESS	START	FINISH	MAN HOURS
Measuring			
TOTAL			

STANDARD (unit mm)	Clearance limit
12V140 -3 = Ø 52.000 - 51.994	0.10

QUANTITY	
----------	--

INSPECTION MEASURING PROCESS												
CYL.NO.		CHECK POSITION	1			2			3			Clearance actual
			MAC	QA	SPV	MAC	QA	SPV	MAC	QA	SPV	
RIGHT BANK	1	X - X'										
		Y - Y'										
	2	X - X'										
		Y - Y'										
	3	X - X'										
		Y - Y'										
	4	X - X'										
		Y - Y'										
	5	X - X'										
		Y - Y'										
	6	X - X'										
		Y - Y'										
LEFT BANK	1	X - X'										
		Y - Y'										
	2	X - X'										
		Y - Y'										
	3	X - X'										
		Y - Y'										
	4	X - X'										
		Y - Y'										
	5	X - X'										
		Y - Y'										
	6	X - X'										
		Y - Y'										

INSPECTION RESULT :

INSPECTION BY :

INSPECTION & APPROVED BY :

MECHANIC(MECH)

QUALITY ASSURANCE (QA)

PROD. SUPERVISOR (SPV)



PT SAPTAINDRA SEJATI

CYLINDER BLOCK INSPECTION CHECKSHEET

KOMATSU ENGINE 12V140E-3 SERIES

MACHINE MODEL

HD785-7

RO NO :

ENGINE SECTION

CYLINDER BLOCK INSPECTION SHEET ENGINE

MACHINE MODEL		CYLINDER BLOCK P/N	6219-21-1100	DOCUMENT NO.	ENG-CB-12V140/3-2-001
ENGINE MODEL	SAA12V140 SERIES	INSPECTION DATE		PUBL. DATE	5-Aug-14
ENGINE NO.		WO NUMBER		REVISION NO.	03
CUSTOMER NAME				PAGE	2 OF 3

✓ : GOOD CONDITION X : BAD CONDITION ☒ : NONE

SKETCH DRAWING OR PHOTOES	NO	ITEM CHECK	INSPECTION		REMARK
			OK	NOT OK	
Cylinder block LH 	1	Counter bore surface area			
	2	Water jacket wall area			
	3	Cylinder block upper surface			
	4	O ring liner area			
		Inspeksi pada semua thread, pengecekan dengan menggunakan bolt yang sesuai spec dengan thread yang ada di cyl block.			TOTAL THREAD 72 THREAD AND 7 PLUG
		# All thread on camfollower side			
		# All thread cyl head&Top surface LH cyl block			
		# All thread on (common rail side cyl block)			
	5	Timing gear hole area			
	6	Oil pump contact area			
	7	Thrust bearing surface			
	8	Main bore front surface			
	9	Oil and water hole gallery surface area			
	10	Blind cover water area			
	11	Camshaft hole surface rear area RH LH			
	12	Remove all plug on oil and water gallery			
		All Thread condition			TOTAL THREAD 48 FRONT SIDE, 40 THREAD REAR SIDE
		Inspeksi pada semua thread, pengecekan dengan menggunakan bolt yang sesuai spec dengan thread yang ada di cyl block.			
		# All thread timing gear shaft & Camshaft			
		# All thread front cover mounting			
	13	Oil cooler hole area			
	14	Oil adapter hole area			
	15	Regulator hole area			
	16	Water hole area			
	17	O ring liner area			
	18	Counter bore surface area			
	19	Water jacket wall area			
	20	Cylinder block upper surface			
	21	All Thread condition			TOTAL THREAD 112 THREAD
		Inspeksi pada semua thread, pengecekan dengan menggunakan bolt yang sesuai spec dengan thread yang ada di cyl block.			
		# All thread Camfollower & Oil cooler area			
		# All thread oil adapter and under oil cooler			
		# All thread cyl head & all thread on surface RH			
	22	Cylinder block blind cover area			
	23	Retap thread on front and rear hanger			
	24	Thread mounting piston cooling nozzle			
	25	All Main metal upper seat area			TOTAL THREAD UPPER 26 THREAD, LOWER 72 THREAD
		Inspeksi pada semua thread, pengecekan dengan menggunakan bolt yang sesuai spec dengan thread yang ada di cyl block.			
		# All thread cyl block hanger and blind cover area			
		# All thread oil pan, main cap & cooling nozzle mounting			

Notes :

1. Semua thread yang tertutup dengan tanah (soil) harus dilakukan retap dan cleaning oleh man power disassy.

Checked by,

Inspected by,

Approved by,

(MECHANIC)

(QA INSPECTOR)

(PROD.SUPERVISOR)



PT SAPTAINDRA SEJATI

ENGINE ASSY

SHORTBLOCK ASSEMBLY

CHECKSHEET

KOMATSU ENGINE 12V140E-3 SERIES

MACHINE MODEL

HD785-7

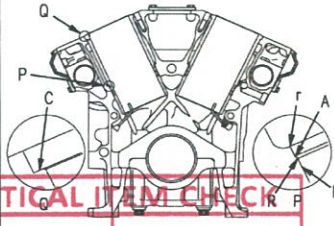
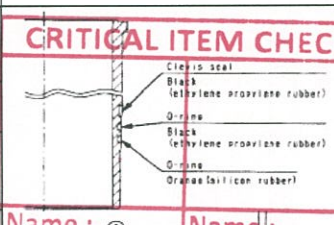
RO NO :

ENGINE SECTION

ASSEMBLY CHECK SHEET ENGINE ASS'Y KOMATSU ENGINE 12V140E-3	WO NUMBER		DOCUMENT NO	QR03-ENG-011
	ENGINE MODEL		PUBLIC.DATE	28/08/2009
	ENGINE S/N (new)		REVISION NO.	06
	MACH.MODEL		PAGE	1 / 6

PROCESS	START/DATE&TIME		ASSEMBLY BY	Short block	
	FINISH/DATE&TIME			Main Line	

Firing Order	1R-1L-5R-5L-3R-3L-6R-6L-2R-2L-4R-4L		All dimensions in millimetres
Valve Clearance	Intake	0.35 ± 0.02	All torques in Kilogram metres
	Exhaust	0.57 ± 0.02	

No.	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN			SKETCH DRAWING or PHOTOS
					MEC	GL	SPV	
1	Cylinder Block	Bores	Check that the liner counter bore is clean and serviceable prior to fitting liner.					 CRITICAL ITEM CHECK Name : QA INSPECTOR Name : PROD. SUPERVISOR
		Inside of oil hole	Check that all oil holes are free from foreign matter such as chips or dirt.					
		U-plugs & blind plugs	Make sure that all U-plugs and blind plugs have been installed.					
			Make sure that Cyl. block condition is ready to used completely - Tightening plug : Hexagon plug 5 mm Act : kgm	1.0 ~ 2.0 kgm (Komatsu assembly manual)				
			• Oversize head gasket required: Yes No	THE OTHERS PLUG, SEE PLUG TABLE FOR TIGHTEN				
		S/N	Cylinder block S/N new: Cylinder block S/N original:					
		Liner deck portion	See that three bondLG-6 has been applied to the proper area.					
2	Cylinder Liner	Crevice seal O-rings	Check that the chamfered side is facing down. Check that the colour of the bottom O-ring is orange					 CRITICAL ITEM CHECK Name : QA INSPECTOR Name : PROD. SUPERVISOR
		Liner	Apply oil immediately before installing. Set the "T" mark on the top surface of the liner facing the front. Using a cloth, wipe off dirt and oil from contact surface of liner lip and cylinder block counterbore.					
		Liner Protrusion	Check the protrusion of the liner: R bank L bank	0.10~0.15mm (Shop Manual Page. 50-5)				
3	Crank shaft	Crank Main Bearings	Make sure that stamp mark on Crank Shaft is available. Check Code of Main metal, code are: STD/0.25/0.50 Code: STD 0.25 0.50	STD or US 0.25 US 0.50				 CRITICAL ITEM CHECK Name : QA INSPECTOR Name : PROD. SUPERVISOR
			before tightening. Coat the inside face with engine oil.					

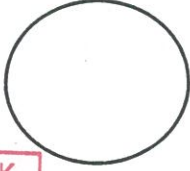
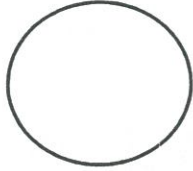
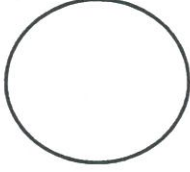
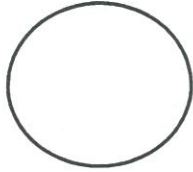
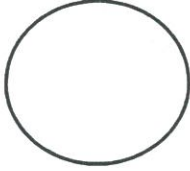
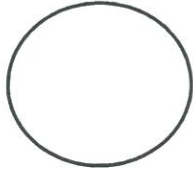
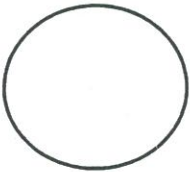
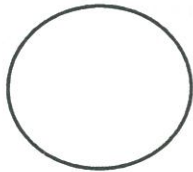
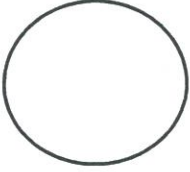
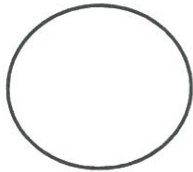
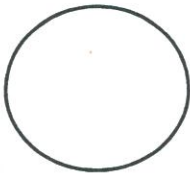
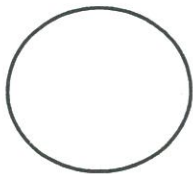
MEASUREMENT LINER PROTRUSION EACH CYLINDER

FRONT ENGINE



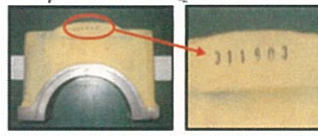
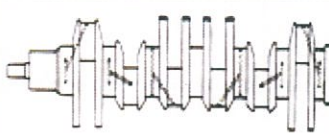
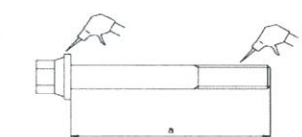
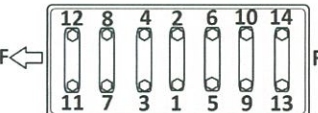
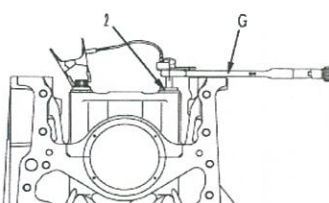
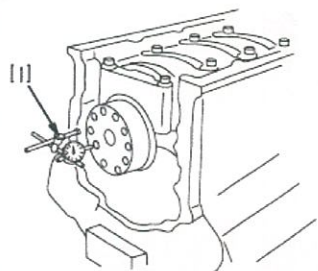
LH

RH



CRITICAL ITEM CHECK	
Name :	Name :
QA INSPECTOR	PROD. SUPERVISOR

ASSEMBLY CHECK SHEET ENGINE ASS'Y KOMATSU ENGINE 12V140E-3	WO NUMBER		DOCUMENT NO	QR03-ENG-011
	ENGINE MODEL		PUBLIC.DATE	28/08/2009
	ENGINE S/N (new)		REVISION NO.	06
	MACH.MODEL		PAGE	2 / 6

			Do no coat the rear surface of the bearing • Upper bearing has oil hole • Lower bearing has no oil hole • Protrusion of Dowel Pin Act : mm *Make sure the number of main cap same with S/N Engine Block. CRANKSHAFT S/N: (Serial No Engine at rear face of Crankshaft)	2.7~3.4 mm (Shop Manual Page. 50-5)			 
4	Crank shaft Cont.	Thrust bearing	Install the thrust bearing with the groove on the crankshaft side.				
		Torque of Bearing Cap	Bolts must be oiled. When tightening, start from the center and work in order to the Outside Using limit length (a) of bolt stem: CRITICAL ITEM CHECK Torque sequence mounting bolt main cap. Name : Name : QA INSPECTOR PROD. SUPERVISOR 1st step : Act : kgm 29 ± 1.5 Kgm 2nd step: Act : kgm 58 ± 1.0 Kgm 3rd step : Act : ° 90 ~120 ° (Shop Manual Page. 50-7) * After tightening bolt, make 1 punch mark on the head of each bolt. (Don't make a punch mark when using a new bolt) * Punch mark max. 5 punch marks.	Less than 176.3 mm (Shop Manual Page. 50-6)		 	
		Seal Surface	Clean of all dirt and dust on the contact surface of the front and rear seals				
		End Play	Measure crankshaft End Play with dial gauge Act : mm • Tolerance: Check that shaft turns smoothly • Standard rotating force: Act : kgm	0.140 -0.320 mm (Shop Manual Page. 50-7) 1.5 Kgm (Shop Manual Page. 50-7)			

ASSEMBLY CHECK SHEET ENGINE ASS'Y KOMATSU ENGINE 12V140E-3	WO NUMBER		DOCUMENT NO	QR03-ENG-011
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			Radial out & Face Runout Crankshaft	Measure Face & Radial out of Crankshaft at Front and Rear surface Crankshaft before install Con rod assembly.	Maximum 0.05 mm (std alignment main bearing bore)					<table border="1"> <tr><td>Front Face Runout:</td><td>mm</td></tr> <tr><td>Front Radial out:</td><td>mm</td></tr> <tr><td>Rear Face Runout:</td><td>mm</td></tr> <tr><td>Rear Radial out:</td><td>mm</td></tr> </table>	Front Face Runout:	mm	Front Radial out:	mm	Rear Face Runout:	mm	Rear Radial out:	mm													
Front Face Runout:	mm																														
Front Radial out:	mm																														
Rear Face Runout:	mm																														
Rear Radial out:	mm																														
5	Piston and Con. rods		Piston	When assembling a new connecting rod, mark the cylinder number with an electric punch. Do not make a stamped mark. Set up each ring with the stamped mark facing up. Note the position of the ring gap. For the oil ring, take out the expander and install it into the piston, then install the oil ring. Confirm that the expander is properly fitted to the groove.	top ring: 1RS second ring: 2RX					 																					
			Snap Ring	Assemble piston and connecting rod. Confirm that the snap ring which secures the piston pin is properly fitted to the groove.																											
			Conrod Bearing	Check that the connecting rod moves smoothly -Check size of Pin metal and recheck Code Pin metal on rear surface of metal. (Code of Pin Metal are: STD/0.25/0.50). Code of Pin metal: <table border="1"> <tr> <td>STD</td> <td>0.25</td> <td>0.50</td> </tr> <tr> <td></td> <td></td> <td></td> </tr> </table>	STD	0.25	0.50				STD or US 0.25 US 0.50					CRITICAL ITEM CHECK Name : Name : QA INSPECTOR PROD. SUPERVISOR															
STD	0.25	0.50																													
No.	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN	MEC	GL	SPV	SKETCH DRAWING or PHOTOES																						
6	Piston and Con. rods con	Con. rod Bearing	Match the number of con. Rod are aligned Check that there is no dirt at the rear face of the bearing before assembling. <table border="1"> <tr> <td>CYL</td> <td>1</td> <td>2</td> <td>3</td> <td>4</td> <td>5</td> <td>6</td> </tr> <tr> <td>UP</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>LOW</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> </table> Do not coat the rear surface bearing with oil.	CYL	1	2	3	4	5	6	UP							LOW													
CYL	1	2	3	4	5	6																									
UP																															
LOW																															
			Inserting Piston & Con. Rod	Check that the hole in the upper bearing and the oil hole in the co. rod are aligned. Set the cylinder block on its side so that the assembly does not enter the cylinder suddenly when inserted.																											

ASSEMBLY CHECK SHEET ENGINE ASS'Y KOMATSU ENGINE 12V140E-3	WO NUMBER		DOCUMENT NO	QR03-ENG-011
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	MACH.MODEL		PAGE	4 / 6

Coat the cylinder liner with engine oil (SAE 15W-40) uniformly around the whole circumference by Hand.

Check that number of piston and connecting rod assembly match cylinder number.

Coat connecting rod lower bearing with engine oil, spreading over whole part with finger.

Con rod bolts are to have oil (SAE15W-40) on the threads and under the washers.

Big End Torque

1st step: Act : kgm **7 ~ 8 kgm**

2nd step: Act : kgm **7 ~ 8 kgm**

3rd step: Act : ° **90° ~ 120°**
(Shop Manual Page. 50-11)

After tighten of bolt, make 1 punch mark on the head of each bolt (Don't make a punch mark when using a new bolt)
* Punch mark max. 5 punch marks.

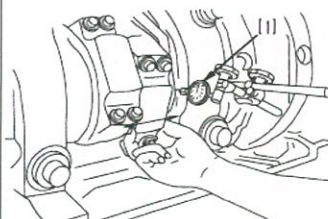
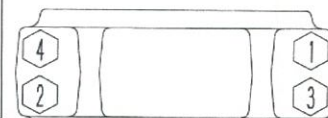
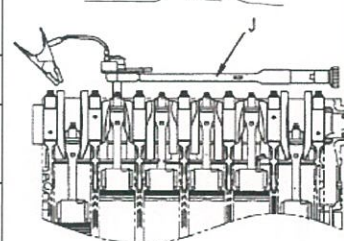
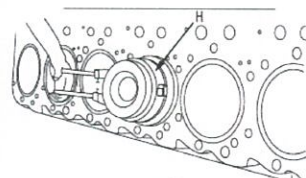
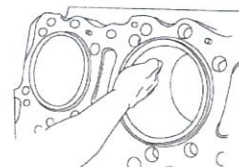
Measure con rod side clearance

0.300 ~ 0.454 mm
(Shop Manual Page. 50-11)

CRITICAL ITEM CHECK

Name :
QA INSPECTOR

Name :
PROD. SUPERVISOR



No.	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN			SKETCH DRAWING or PHOTOES
					MEC	GL	SPV	
7	Piston and Con. rods cont.		Recheck/pastikan bahwa posisi piston sudah benar (dilihat dari posisi top). * <u>Cowakan yang lebih dalam ada di sisi intake</u> * <u>Cowakan yang dangkal ada di sisi exhaust</u>					

CRITICAL ITEM CHECK

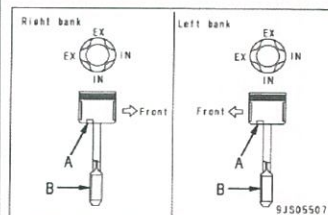
CYL 1 2 3 4 5 6

RH

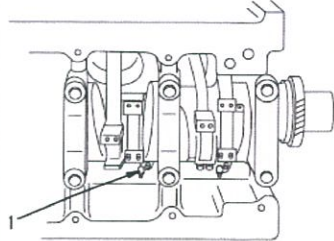
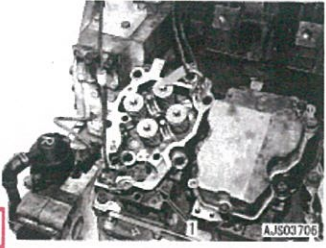
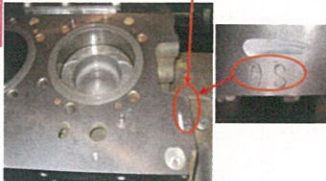
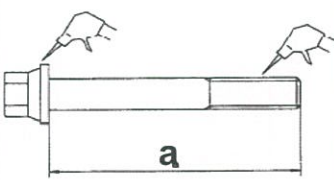
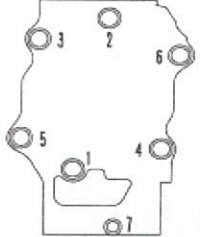
LH

Name :
QA INSPECTOR

Name :
PROD. SUPERVISOR



ASSEMBLY CHECK SHEET ENGINE ASS'Y KOMATSU ENGINE 12V140E-3	WO NUMBER		DOCUMENT NO	QR03-ENG-011
	ENGINE MODEL		PUBLIC DATE	28/08/2009
	ENGINE S/N (new)		REVISION NO.	06
	MACH. MODEL		PAGE	5 / 6

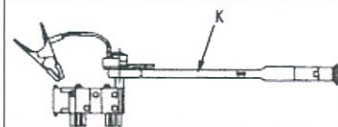
	Piston Cooling Nozzle	Check that the piston cooling nozzle is not clogged and that the angle is correct and Install to Block. Tightening torque Act : _____ kgm	5.5 ± 2.0 Kgm (Shop manual Page. 50-11)																									
8	Cylinder Head	Grommets - Check that there is no dust or dirt inside the cylinder and on mounting surface. - When installing the gasket, check that the grommet has been fitted and has not come - The grommets in the head gasket may become deformed by swelling, so do not coat them with gasket sealant. - Check and mark grommet with white marker. <table border="1"> <tr> <td>Cyl.</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr> <tr> <td>RH</td><td></td><td></td><td></td><td></td><td></td><td></td></tr> <tr> <td>LH</td><td></td><td></td><td></td><td></td><td></td><td></td></tr> </table>	Cyl.	1	2	3	4	5	6	RH							LH											 Cylinder head assembly. 30 kg
Cyl.	1	2	3	4	5	6																						
RH																												
LH																												
	O/S head gasket	If oversize head gaskets are to be fitted, the block must be stamped O/S. <table border="1"> <tr> <td>STD</td><td>OS</td></tr> <tr> <td></td><td></td></tr> </table>	STD	OS			Name : _____ STD or OVER QA SIZE INSPECTOR Name : _____ PROD. SUPERVISOR				 CODE FOR OVSIZE GASKET																	
STD	OS																											
	Bolts	-The head bolts must have no dirt in the threads or under the washer, or in the threaded holes in the block. -Coat the thread and bolt head with Oil. The threaded holes in the block must be free from dirt. A tap can be used to clean the threads	limit length (a) : Short bolt = Less than 170.8 mm Long bolt = Less than 205.8 mm																									
No.	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN			SKETCH DRAWING or PHOTOS																				
					MEC	GL	SPV																					
9	Cylinder Head Cont.	Torque	- Install cyl. head gasket. - Lift cy. Head ass'y then install to the block - Recheck bahwa marking nomor di Cyl. Head housing sesuai dgn urutan Pada Cylinder block. - Tighten Cyl. head mounting bolt . * No.1 - 6 : Step 1st Act : _____ kgm	7 Kgm 14 ~ 16 Kgm																								

ASSEMBLY CHECK SHEET

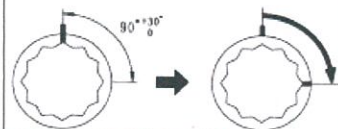
ENGINE ASS'Y

KOMATSU ENGINE 12V140E-3

Step 2nd	Act :	kgm	29 ~ 30 Kgm
Step 3rd	Act :	kgm	90 ~ 120°
Step 4rd	Act :	kgm	(Shop Manual Page. 50-27)



* No.7 : <table border="1" data-bbox="486 562 782 602"> <tr> <td>Act :</td> <td>kgm</td> </tr> </table> torque in the order shown. After tightening, make a punch mark on the bolt head. If the bolt has 5 marks, do not use it. Replace it with new one.	Act :	kgm	6.75 ± 0.75 kgm (Shop Manual Page. 50-27)
Act :	kgm		
Install the water tube. Water tube joint bolts : <table border="1" data-bbox="486 629 782 672"> <tr> <td>Act :</td> <td>kgm</td> </tr> </table>	Act :	kgm	1.0 ~ 1.3 Kgm (Shop Manual Page. 50-28)
Act :	kgm		
Install heat label at cylinder head No.6 <table border="1" data-bbox="486 689 782 730"> <tr> <td>Act :</td> <td>°</td> </tr> </table>	Act :	°	99°C ~ 127°C (210°F ~ 260°F)
Act :	°		



CRITICAL ITEM CHECK	
Name :	Name :
QA INSPECTOR	PROD. SUPERV SCR

ASSEMBLY BY

INSPECTED & APPROVED BY

MECHANIC (MEC)

QUALITY ASSURANCE (QA)

GL/SPV PRODUCTION



PT SAPTAINDRA SEJATI

CYLINDER BLOCK

DELIVERRY INSPECTION

CHECKSHEET

KOMATSU ENGINE 12V140E-3 SERIES

MACHINE MODEL

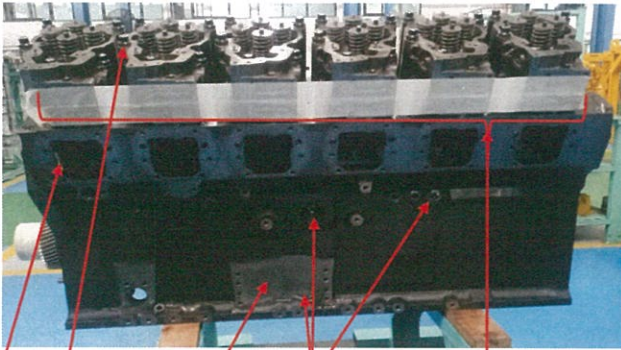
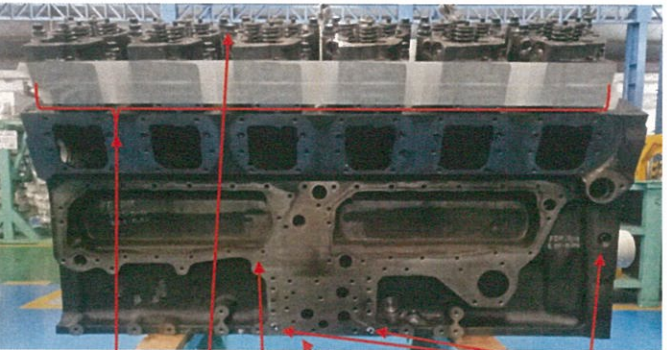
HD785-7

RO NO :

ENGINE SECTION

DELIVERY INSPECTION SHORT BLOCK ENGINE CHECKSHEET	DOCUMENT NO.	RC/QR06/EG/01/01
	PUBL. DATE	
	REVISION NO.	
	PAGE	1

JOB NUMBER		PART NAME	ENGINE ASS'Y
ENGINE MODEL	SAA12V140E-3	DATE / TIME	
ENGINE SER. NO			
MACHINE MODEL	HD 785-7		

SKETCH DRAWING OR PHOTOES	
LEFT SIDE 	RIGHT SIDE 

✓ : GOOD CONDITION X : BAD CONDITION ⊗ : NONE

LEFT				RIGHT			
NO	ITEM	✓	CONDITION	NO	ITEM	✓	CONDITION
1	Bushing camshaft			7	Bushing camshaft		
	-hole lubricating align looks 60%				-hole lubricating align looks 60%		
	-No scratch and damage				-No scratch and damage		
	-Align bushing camshaft(special tool)				-Align bushing camshaft(special tool)		
	-Conection bushing up side position				-Conection bushing up side position		
2	Hexagon plug			8	Hexagon plug		
3	Bolt cyl head(after +90° turning angle)			9	Bolt cyl head(after +90° turning angle)		
4	Intake hole should be closed(Tape)			10	Intake hole should be closed(Tape)		
5	Surface mounting all component no corotion			11	Surface mounting all component no corotion		
6	All tread bolt id good condition			12	Hexagon and blind plug under cyl block.		

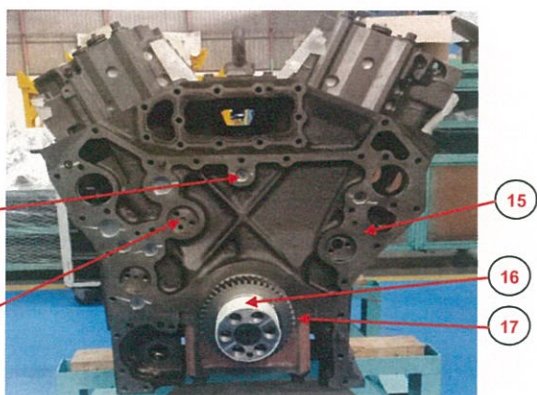
DELIVERY INSPECTION SHORT BLOCK ENGINE CHECKSHEET

DOCUMENT NO.	RC/QR06/EG/01/01
PUBL. DATE	
REVISION NO.	
PAGE	2

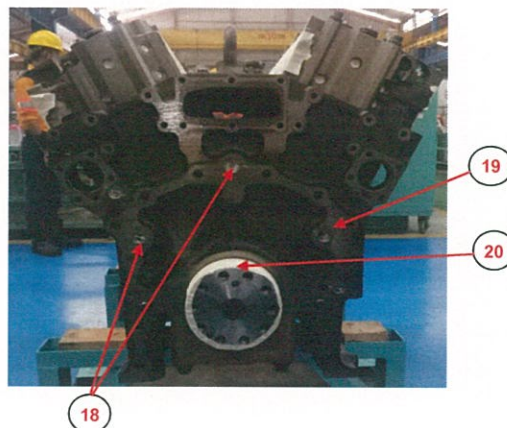
JOB NUMBER		PART NAME	ENGINE ASS'Y
ENGINE MODEL	SAA12V140E-3	DATE / TIME	
ENGINE SER. NO			
MACHINE MODEL	HD 785-7		

SKETCH DRAWING OR PHOTOES

FRONT



REAR



✓ : GOOD CONDITION X : BAD CONDITION ⊗ : NONE

FRONT				REAR			
NO	ITEM	✓	CONDITION	NO	ITEM	✓	CONDITION
13	Timing gear hole(no damage and corrotion)			18	Blind & Hexagon plug		
14	Blind & Hexagon plug			19	Surface for mounting all component no damage&corotion		
15	Surface for mounting all component no damage&corotion			20	Rear seal surface crankshaft (should be closed tape)		
16	Frontseal surface crankshaft (should be closed tape)						
17	Gear crankshaft not scratch and pitcing						

DELIVERY BY SHORT BLOCK SECTION

DELIVERED BY :	
MACHINIST NAME :	
SIGN	

RECEIVING INSPECTION BY MAINLINE SECTION

RECEIVED DATE :	
MECHANIC NAME :	
SIGN	



PT SAPTAINDRA SEJATI

ENGINE ASSY

DELIVERRY INSPECTION

CHECKSHEET

KOMATSU ENGINE 12V140E-3 SERIES

MACHINE MODEL

HD785-7

RO NO :

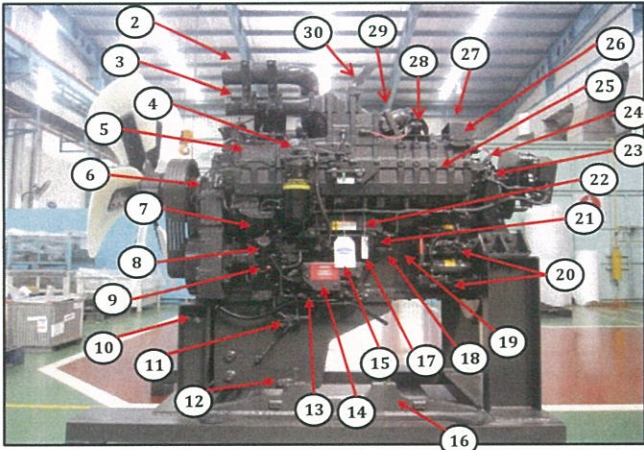
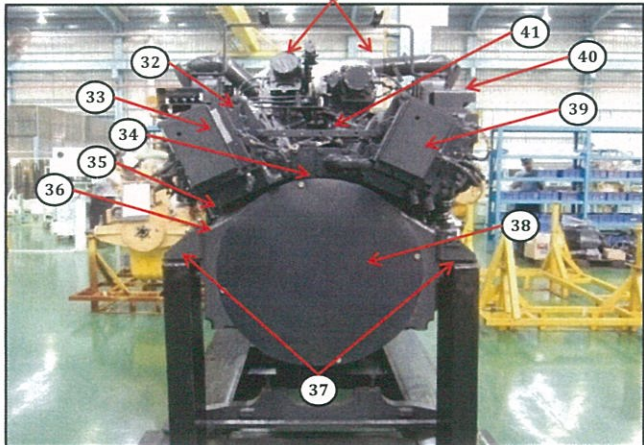
ENGINE SECTION

DELIVERY INSPECTION SHEET ENGINE

MACHINE MODEL	HD785-7	ENGINE PART NO.	6219-B0-0031	DOCUMENT NO.	RC/QR06/EG/24/01
ENGINE MODEL	SAA12V140E-3	WO ORDER NO.		PUBL. DATE	7-Sep-10
ENGINE NO.		DELIVERY DATE		REVISION NO.	01
CUSTOMER NAME				PAGE	1 OF 3

BELOW ARE THE ITEMS RELATED TO DELIVERY INSPECTION AND WHILE YOU DOING THIS INSPECTION JOB, PLEASE REFERRING TO THE REMANUFACTURED COMPONENTS - STANDARD CONFIGURATION.

✓ : GOOD CONDITION X : BAD CONDITION ⊗ : NONE

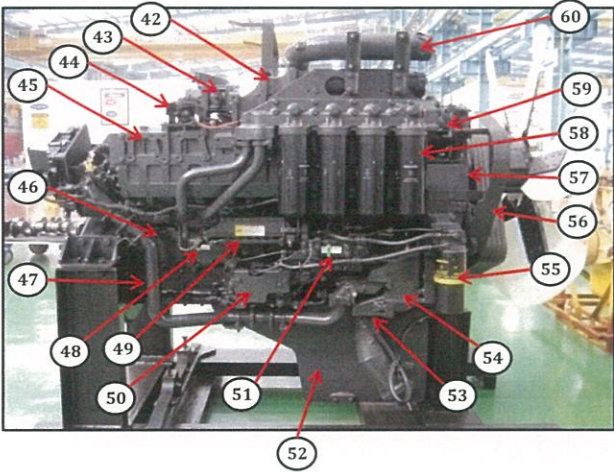
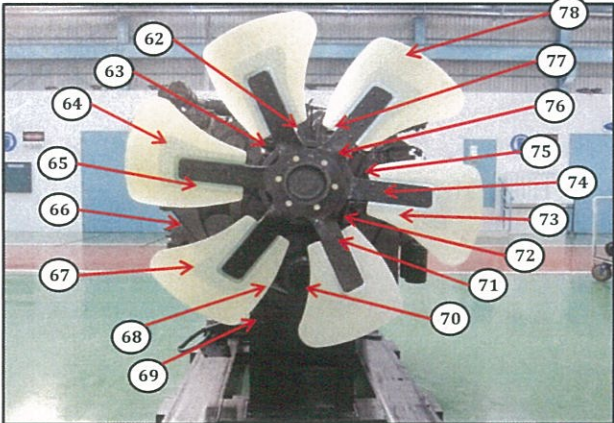
SKETCH DRAWING OR PHOTOES	NO.	ITEM	INSPECTION CONDITION
Left Side View 	1	Painting (Y/B)	
	2	Tube Air Intake Connection LH	
	3	Air Connector LH	
	4	Fuel Filter Assy LH	
	5	Engine Name Plate	
	6	Tube, Breather With Hose LH	
	7	Fuel Supply Pump Assy LH (P/N : 6219-71-1110)	
	8	Tube Oil Filler Cap	
	9	Gauge Oil (STD : H+ 0~10 mm)	
	10	Engine Stand (Original / Std)	
	11	Oil Level Sensor Assy	
	12	Drain Oil Plug	
	13	Priming Pump Assy With Cover LH	
	14	Warning Sticker Oil Level	
	15	Bracket Mounting	
	16	Quality passed Inspection Tag	
Rear Side View 	17	Caution Tag	
	18	Oil Temperature Sensor	
	19	Oil Pressure Sensor	
	20	Starting Motor Assy (7.5 KW) (P/N : 600-813-7542)	
	21	Fuel Pressure Sensor With Limiter Assy LH	
	22	Common Rail Assy	
	23	Boost Pressure Sensor	
	24	Intake Temperature Sensor	
	25	Air Intake Manifold LH	
	26	Exhaust Temperature Sensor (4 pcs) With Bracket	
	27	Customer Responsibilities Sticker	
	28	Heater Switch Assy With Cover LH	
	29	Exhaust Brake Assy LH	
	30	Bracket Exhaust Brake Mounting	
	31	Turbocharger Assy RH P/N : 6505-67-5030	
	32	Heat Tab position	
	33	Engine Controller Assy LH	
	34	Caution Flywheel Sticker	
	35	Pick Up Revolution Sensor	
	36	Flywheel Housing	
	37	Rear Bracket Mounting (2 pcs)	
	38	Flywheel (P/N : 6219-31-4110)	
	39	Engine Controller Assy RH	
	40	Stiker Attention Installation Report	
	41	Exhaust Manifold Assy RH & LH	

DELIVERY INSPECTION SHEET ENGINE

MACHINE MODEL	HD785-7	ENGINE PART NO.	6219-B0-0031	DOCUMENT NO.	RC/QR06/EG/24/01
ENGINE MODEL	SAA12V140E-3	WO ORDER NO.		PUBL. DATE	7-Sep-10
ENGINE NO.		DELIVERY DATE		REVISION NO.	01
CUSTOMER NAME				PAGE	2 OF 3

BELOW ARE THE ITEMS RELATED TO DELIVERY INSPECTION AND WHILE YOU DOING THIS INSPECTION JOB, PLEASE REFERRING TO THE REMANUFACTURED COMPONENTS - STANDARD CONFIGURATION.

✓ : GOOD CONDITION X : BAD CONDITION ⊗ : NONE

SKETCH DRAWING OR PHOTOES	NO.	ITEM	INSPECTION CONDITION
Right Side View 	42	Air Connector RH	
	43	Exhaust Brake Assy RH	
	44	Heater Switch Assy With Cover RH	
	45	Air Intake Manifold RH	
	46	Oil Cooler Assy	
	47	Tube Cooling System Piping	
	48	Fuel Pressure Sensor With Limiter Assy RH	
	49	Common Rail Assy	
	50	Priming Pump Assy With Cover RH	
	51	Fuel Supply Pump Assy RH (P/N : 6219-71-1120)	
		S/N :	
	52	Oil Pan	
	53	Water Inlet Cover Plate	
	54	Water Pump Assy	
	55	Fuel Filter Assy	
	56	Cover Safety Guard Alternator	
	57	Tube, Breather With Hose RH	
	58	Oil Filter Assy	
	59	Alternator Assy (90A) (P/N : 600-825-9330)	
		S/N :	
Front Side View 	60	Tube Air Intake Connection RH	
	62	Plate Front Hanger	
	63	Fan Drive With Pulley Assy	
	64	Block Fuel Return With Sensor Assy RH & LH	
	65	V-Belt Alternator	
	66	Alternator Drive Pulley	
	67	Tension Pulley Assy	
	68	Vibration Damper Assy (P/N : 6215-31-8200)	
	69	Front Support	
	70	Crank Pulley Assy	
	71	Pointer	
	72	Yoke	
	73	V-Belt Fan Set	
	74	Blowbay Sensor Assy	
	75	Water Temperature Sensor	
	76	Thermostat Housing	
	77	Tube Water Outlet (2 pcs)	
	78	Cooling Fan	
	79	RFID Tag, Traceability mounting bolt	
	80	Masking condition	
	81	Check Mounting Bolt (Rear Bracket & Front Support)	
		Std Tightening Torque : 25 kgm	

Notes :

Inspected by,

Approved by,



PT SAPTAINDRA SEJATI

ENGINE ASSY

MAINLINE ASSEMBLY

CHECKSHEET

KOMATSU ENGINE 12V140E-3 SERIES

MACHINE MODEL

HD785-7

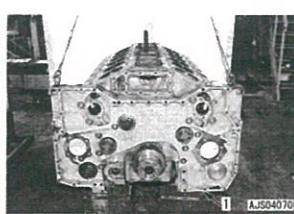
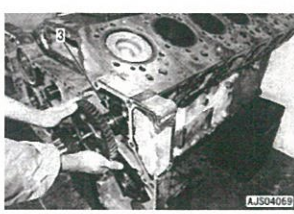
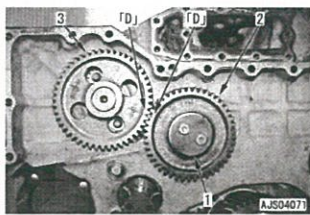
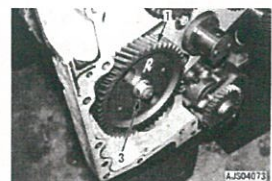
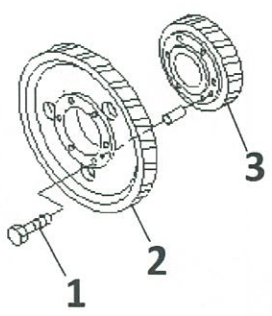
RO NO :

ENGINE SECTION

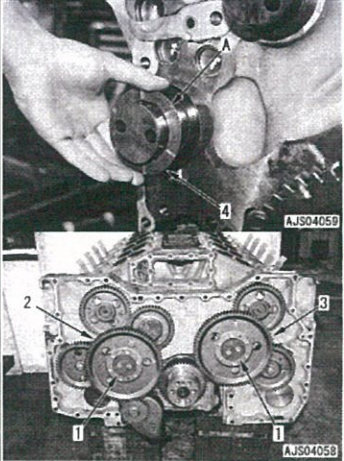
MAIN ASSEMBLY CHECK SHEET ENGINE ASS'Y KOMATSU ENGINE 12V140E-3	WO NUMBER		DOCUMENT NO	RC/QA/EG/014
	ENGINE MODEL	SAA12V140-3	PUBLIC DATE	28/08/2009
	ENGINE S/N (new)		REVISION NO.	06
	MACH.MODEL	HD 785-7	PAGE	1 / 12

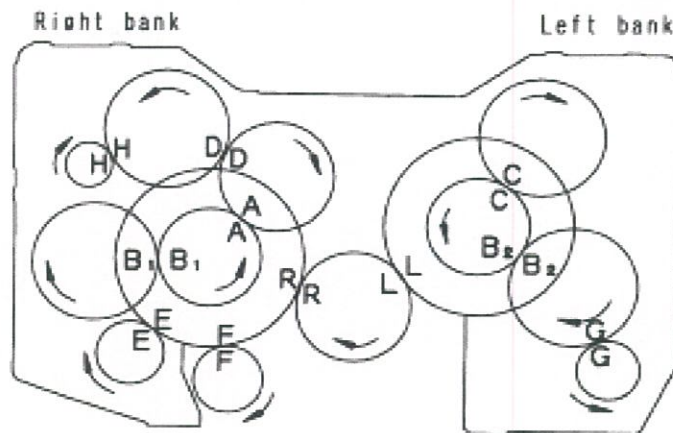
PROCESS	START/DATE&TIME		ASSEMBLY BY	Short block	
	FINISH/DATE&TIME			Main Line	

Firing Order	1R-1L-5R-5L-3R-3L-6R-6L-2R-2L-4R-4L		All dimensions in millimetres
Valve Clearance	Intake	0.35 ± 0.02	All torques in Kilogram metres
	Exhaust	0.57 ± 0.02	

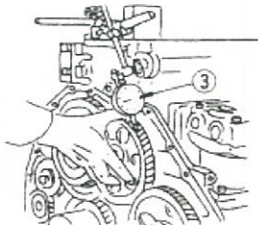
No.	PARTS	CHECK	CONDITIONS TO BE CHECKED	SPEC.	SIGN			SKETCH DRAWING or PHOTOS
					MEC	GL	SPV	
1	Timing Gear Case		- Prior to installation, fix several portions on gasket temporarily using the gasket sealant so that it may be displaced. - Mounting Bolt: Act : kgm	10.0 ~ 12.5 kgm (Shop Manual page. 12-06)				
2	Camshaft		- Turn the cam gear and tighten the mounting bolt, through lightening hole. * Replace the current mounting bolts with a new one - Cam shaft mounting bolt : Act : kgm	9.0 ± 3.5 Kgm (Shop Manual page. 50-12)				
3	Idler gears		- When install idler gear, align the counter mark "D" to that the cam gear for the right bank. - Coat the idler gear bushing inner surface with engine oil (15W40). - Install plate (1) and tighten mounting bolts. - Tightening torque : Act : kgm	10.0 ~ 12.5 kgm (Shop Manual Page. 50-14)				
			- Install key (2) to the shaft - Install supply pump gear and tighten mounting nut. *Coat on thread of shaft supply pump use adhesive Locktite LT2(High strength type) - Tightening torque mounting nut : Act : kgm	18 ~ 20 kgm (Shop Manual Page. 50-14)				
			- Install pin dowel at Idle gear (small)(3), - Install idle gear (large)(2) to idle gear (small), match pin dowel idle gear (small)(3) to hole idle gear (large)(2). - Mounting bolt (1) tighten: Act : kgm	6.0~7.5 Kgm				

MAIN ASSEMBLY CHECK SHEET ENGINE ASS'Y KOMATSU ENGINE 12V140E-3	WO NUMBER		DOCUMENT NO	RC/QA/EG/014
	ENGINE MODEL	SAA12V140-3	PUBLIC DATE	28/08/2009
	ENGINE S/N (new)		REVISION NO.	06
	MACH. MODEL	HD 785-7	PAGE	2 / 12

No.	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN			SKETCH DRAWING or PHOTOS
					MEC	GL	SPV	
4	Idler Gears Cont.	Idler gear (large)	- Set the No.1 cylinder head on the right bank to the top dead center position. - Coat the idler gear bushing inner surface with engine oil (15W40). - Install trust plate (4) to the shaft. - Install idler gear (2) and (3), then tighten mounting bolts. - Tightening torque mounting bolts : Act : kgm *Install gear such the counter marks as well as that of respective gear may come to the positions shown on the figure.	10.0 ~ 12.5 kgm (Shop Manual Page. 50-15)				



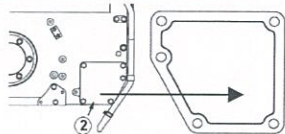
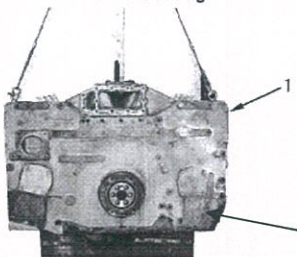
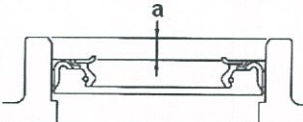
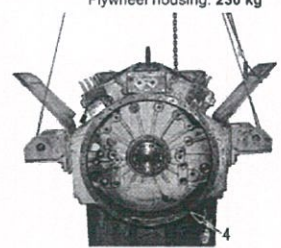
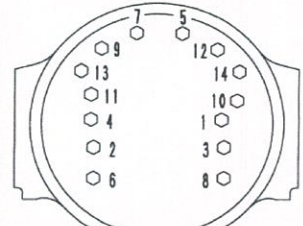
CRITICAL ITEM CHECK	
Name :	Name :
QA INSPECTOR	PROD. SUPERVISOR

Timing Gear Marked	Check Point	Backlash (mm)		SIGN			SKETCH DRAWING or PHOTOS
		Standard	Measured	MEC	GL	SPV	
A • A	Right idler gear (small) and right sub idler gear.	0.130 ~ 0.390 (0.125 ~ 0.363)					Note : The standard value enclosed in () indicates the backlash between “ spur gears” 
B1 • B1	Right sub idler and right supply pump gear.	0.051 ~ 0.469 (0.125 ~ 0.363)					
B2 • B2	Left sub idler and right supply pump gear.	0.051 ~ 0.469 (0.125 ~ 0.363)					
C • C	Left sub idler gear and left cam gear	0.129 ~ 0.391 (0.125 ~ 0.363)					
D • D	Right idler gear (small) and right cam gear.	0.129 ~ 0.391 (0.125 ~ 0.363)					
E • E	Right idler gear (large) and water pump drive gear.	0.052 ~ 0.481					
F • F	Right idler gear (large) and oil pump gear.	0.137 ~ 0.421					
G • G	Left supply pump gear and compressor gear	0.106 ~ 0.381 (0.138 ~ 0.354)					
H • H	Right cam gear and accessory drive gear	0.085 ~ 0.492 (0.138 ~ 0.354)					
L • L	Crank gear and left idler gear (large)	0.141 ~ 0.425					
R • R	Crank gear and right idler gear (LARGE)	0.141 ~ 0.425					
Check Point		End Play (mm)		SIGN			CRITICAL ITEM CHECK

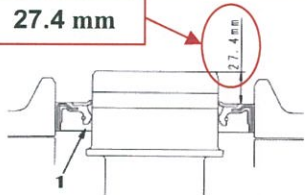
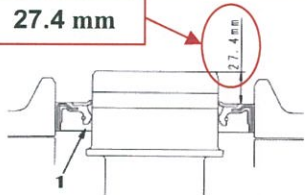
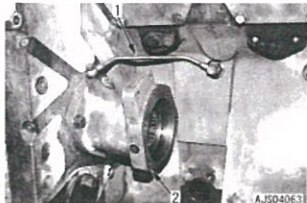
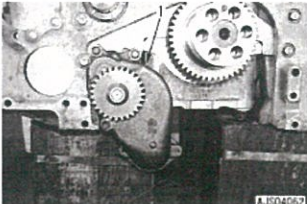
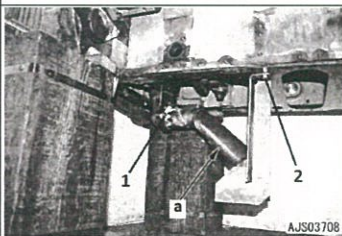
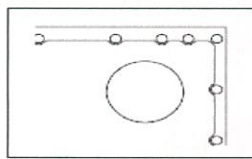
Name :	Name :
QA INSPECTOR	PROD. SUPERVISOR

MAIN ASSEMBLY CHECK SHEET ENGINE ASS'Y KOMATSU ENGINE 12V140E-3	WO NUMBER		DOCUMENT NO	RC/QA/EG/014
	ENGINE MODEL	SAA12V140-3	PUBLIC. DATE	28/08/2009
	ENGINE S/N (new)		REVISION NO.	06
	MACH. MODEL	HD 785-7	PAGE	3 / 12

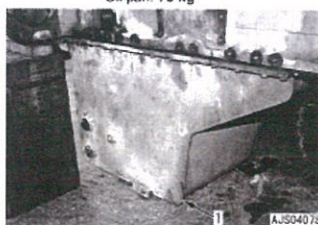
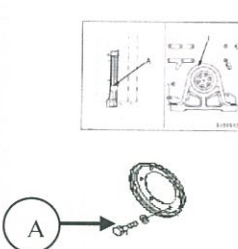
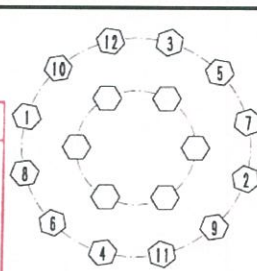
Check Point		End Play (mm)		SIGN		
		Standard	Measured	MEC	GL	SPV
1	Idler gear (large)(LH) end play	0.07~ 0.18				
2	Idler gear (large)(RH) end play	0.07~ 0.18				
3	Idler gear (small) end play	0.07~ 0.18				
4	Supply pump gear(LH) end play	0.07~ 0.20				
5	Supply pump gear(RH) end play	0.07~ 0.20				
6	Oil pump gear end play	0.03 ~0.088				
7	Accessory drive gear end play	0.1 ~0.4				
8	Crank gear end play	0.14 ~ 0.032				
9	Cam gear(LH) end play	0.10 ~ 0.25				
10	Cam gear(RH) end play	0.10 ~ 0.25				

No.	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN			SKETCH DRAWING or PHOTOES													
					MEC	GL	SPV														
5	Gear Case Cover		- Fit gasket, install gear case cover. Torque mounting bolt: Act: kgm * When cover (2) was removed, apply the gasket sealant in the position shown in the figure before install again. *Use Three bond 1207D spec. 	10 ~ 12.5 kgm				Gear case cover: 75 kg 													
		Front Seal	- Check front seal specification <table border="1"><thead><tr><th>S/W</th><th>STD</th><th>OS</th></tr></thead><tbody><tr><td></td><td></td><td></td></tr></tbody></table> - Press in the front seal uniformly so that it is not at an angle.	S/W	STD	OS				STD or OVER SIZE S = Single Lip Seal W= Double Lip Seal				 <table border="1"><thead><tr><th>Dimension b</th><th>Bolt hole 6-12 x 1.75</th><th>Oil seal press-fitting depth a</th></tr></thead><tbody><tr><td>109 mm</td><td>Absent</td><td>13 – 14 mm</td></tr><tr><td>114 mm</td><td>Present</td><td>18 – 19 mm</td></tr></tbody></table>	Dimension b	Bolt hole 6-12 x 1.75	Oil seal press-fitting depth a	109 mm	Absent	13 – 14 mm	114 mm
S/W	STD	OS																			
Dimension b	Bolt hole 6-12 x 1.75	Oil seal press-fitting depth a																			
109 mm	Absent	13 – 14 mm																			
114 mm	Present	18 – 19 mm																			
6	Flywheel Housing	assembly	- No crack and no damage - No damage thread hole - Coat with gasket sealant(Three bond 1215 spec) then raise housing and install - Coat mounting bolt with engine oil SAE15W-40 - Tighten the flywheel housing mounting bolts in order from diagram. Oil bolt threads.					Flywheel housing: 230 kg 													
		1 st step: Act : kgm	35± 5 kgm																		
		2 nd step:	Loosen it																		
		3 rd step: Act : kgm	42± 2 kgm (Shop Manual Page. 50-20)																		

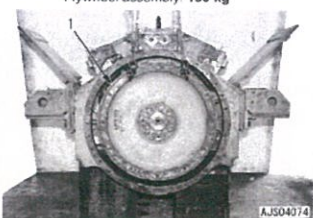
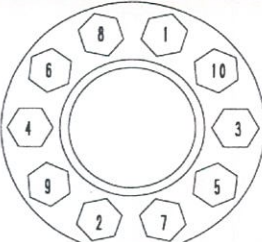
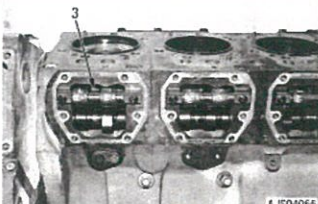
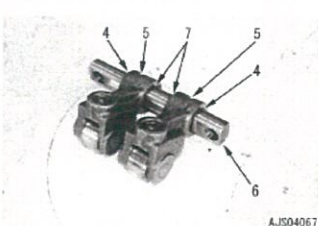
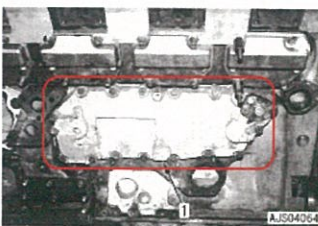
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7		Rear Seal	<table><tr><td>S/W</td><td>STD</td><td>OS</td></tr><tr><td></td><td></td><td></td></tr></table>  - Make sure that the rear seal no pitting, no scratch. - Press fit the rear seal (1)by using installation tool to obtain correct position. Be careful not to damage the seal when installing the housing. STD or OVER SIZE S= Single Lip W= Double Lips <td></td> <td></td> <td></td> <td> 27.4 mm  For New Crank shaft = 20mm </td>	S/W	STD	OS							27.4 mm  For New Crank shaft = 20mm							
S/W	STD	OS																		
No.	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN			SKETCH DRAWING or PHOTOS												
					EC	GL	SPV													
8	Supply pump Drive Case		- Install supply pump drive case Tigthen mounting bolt: LH Bank <table><tr><td>Act</td><td>kgm</td></tr><tr><td></td><td></td></tr></table> RH Bank <table><tr><td>Act</td><td>kgm</td></tr><tr><td></td><td></td></tr></table> - Install oil tube. * Install oil tube on the left bank alone. - Oil tube joint bolts : <table><tr><td>Act :</td><td>kgm</td></tr><tr><td></td><td></td></tr></table>	Act	kgm			Act	kgm			Act :	kgm			6.0~7.5 Kgm 10 ~ 12.5 kgm 0.8 ~ 1.5 Kgm (Shop Manual Page. 50-13)				 AJS04063
Act	kgm																			
Act	kgm																			
Act :	kgm																			
9	Oil Pump Assy	Pump	<table><tr><td>Oil Pump</td><td>New</td><td>Reuse</td></tr><tr><td></td><td></td><td></td></tr></table> - Check all of pump condition - Fill in oil engine to inner of pump - Rotate pump by hand smoothly - Fit oring for sensor side - Mounting oil pump and tighten mounting bolts. <table><tr><td>Act :</td><td>kgm</td></tr><tr><td></td><td></td></tr></table>	Oil Pump	New	Reuse				Act :	kgm			10 ~ 12.5 kgm (Shop Manual Page. 00-32)				 AJS04062  [1]		
Oil Pump	New	Reuse																		
Act :	kgm																			
		Main Relief Valve	- Recheck plug of main relief valve (retighten) <table><tr><td>Relief Valve</td><td>New</td><td>Reuse</td></tr><tr><td></td><td></td><td></td></tr></table>	Relief Valve	New	Reuse														
Relief Valve	New	Reuse																		
		Strainer Suction pipe	Install strainer to oil pump(a),Fit oring and tighten mounting bolt. 1 Act: kgm 2 Act: kgm	10 ~ 12.5 kgm 6.0~7.5 Kgm (Shop Manual Page. 00-32)				 AJS03708												
10	Oil Pan		<table><tr><td>Oil Pan</td><td>New</td><td>Reuse</td></tr><tr><td></td><td></td><td></td></tr></table> - Recheck Condition Of Oil Pan (No crack , No band and No damage) - Install the gasket and push up oil pan	Oil Pan	New	Reuse														
Oil Pan	New	Reuse																		

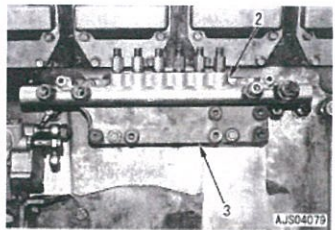
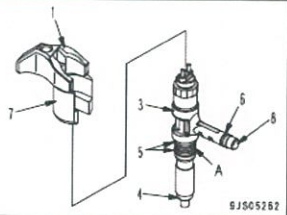
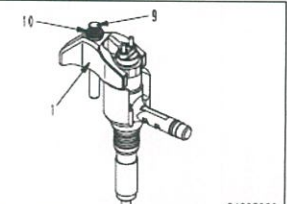
MAIN ASSEMBLY CHECK SHEET ENGINE ASS'Y KOMATSU ENGINE 12V140E-3	WO NUMBER		DOCUMENT NO	RC/QA/EG/014
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	Oil Pan		with a pallet lift, etc. and tighten the mounting bolts - Connect the level sensor connector. - When tightening, start from center and work in order to the out side. - Tighten mounting bolts. Act : kgm	6.0~7.5 Kgm (Shop Manual Page. 00-32)				Oil pan: 75 kg 									
No.	PARTS	CHECK	CONDITIONS TO BE CHECKED	SPEC.	SIGN			SKETCH DRAWING or PHOTOS									
					MEC	GL	SPV										
11	Front Suport		- Prior to the installation, fill the oil hole with Grease. - Volume of grease filled : 20 cc - Tighten mounting bolts cover (A) Act : kgm	STD Grease : (G2-LI) 10 ~ 12.5 Kgm (Shop Manual Page. 00-32)													
12	Dumper and Crank Pulley Ass'y	Damper	- Lift and install damper and crank pulley assembly, * Install the assembly aligning it to the dowel pin of the crankshaft. - Coat mounting bolts and washer with engine oil SAE15W40 - Tightening Mounting bolt: Act : kgm	10 ~ 12.5 kgm (Shop Manual page. 50-19)	CRITICAL ITEM CHECK Name : Name : QA INSPECTOR PROD. SUPERVISOR												
13	Dumper and Crank Pulley Ass'y Cont.	Damper	- Tighten the mounting bolts in 3 steps according to the order indicated in the figure. - Coat mounting bolts and washer with engine oil SAE15W40 - Tightening Mounting bolt: <table><tr><td>Step 1st</td><td>Act :</td><td>kgm</td><td>7.5 ± 2 Kgm</td></tr><tr><td>Step 2nd</td><td>Act :</td><td>kgm</td><td>25 ± 2 Kgm</td></tr><tr><td>Step 3rd</td><td>Act :</td><td>kgm</td><td>76 ± 2 Kgm</td></tr></table>	Step 1st				Act :	kgm	7.5 ± 2 Kgm	Step 2nd	Act :	kgm	25 ± 2 Kgm	Step 3rd	Act :	kgm
Step 1st	Act :	kgm	7.5 ± 2 Kgm														
Step 2nd	Act :	kgm	25 ± 2 Kgm														
Step 3rd	Act :	kgm	76 ± 2 Kgm														

MAIN ASSEMBLY CHECK SHEET ENGINE ASS'Y KOMATSU ENGINE 12V140E-3	WO NUMBER		DOCUMENT NO	RC/QA/EG/014
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14	Flywheel		-Using eye bolt @ (Dia. = 12 mm, Pitch = 1.75 mm), -Sling flywheel , install the flywheel to the crankshaft, -Coat the threads of the bolts and the seat face, with engine oil 15W40 * Make a punch mark on the bolt head each time the bolts are used. Fly wheel mounting bolts : <table><tr><td>Step 1st</td><td>Act :</td><td>kgm</td><td>10 ± 2 Kgm</td></tr><tr><td>Step 2nd</td><td>Act :</td><td>kgm</td><td>30 ± 2 Kgm</td></tr><tr><td>Step 3rd</td><td>Act :</td><td>kgm</td><td>55 ± 2 Kgm</td></tr></table> (Shop Manual page. 50-23)	Step 1st	Act :	kgm	10 ± 2 Kgm	Step 2nd	Act :	kgm	30 ± 2 Kgm	Step 3rd	Act :	kgm	55 ± 2 Kgm				Flywheel assembly: 130 kg  
Step 1st	Act :	kgm	10 ± 2 Kgm																
Step 2nd	Act :	kgm	30 ± 2 Kgm																
Step 3rd	Act :	kgm	55 ± 2 Kgm																
15	Cam Follower	assembly	- Before installing the cam follower ass'y, Wash the oil hole of the mounting bolt. - Make sure that the oil hole is free from dirt. - Use LG-6 liquid gasket on the cover face. (Use Three bond 1215 spec) - Cam follower mounting bolt . <table><tr><td>Act :</td><td>kgm</td></tr></table> - Cam follower cover mounting bolts. <table><tr><td>Act :</td><td>kgm</td></tr></table>	Act :	kgm	Act :	kgm	4.5 ~ 6.0 kgm (Shop Manual page. 50-13) 3.5 ~ 4.5 Kgm (Shop Manual Page. 50-13)				 							
Act :	kgm																		
Act :	kgm																		
	Cam Follower	Torque																	
No.	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN			SKETCH DRAWING or PHOTOS											
					MEC	QA	SPV												
16	Oil Cooler Ass'y		<table><tr><td>Oil cooler</td><td>New</td><td>Reuse</td></tr><tr><td>Front Side</td><td></td><td></td></tr><tr><td>Rear Side</td><td></td><td></td></tr></table> - Check that there is no dust or dirt inside the oil cooler and no crack - Fit gasket and install oil cooler assembly - Oil Cooler mounting bolt: <table><tr><td>Act :</td><td>kgm</td></tr></table> -Use gasket sealant Three bond 1215 spec.	Oil cooler	New	Reuse	Front Side			Rear Side			Act :	kgm	(M10) Bolts : 6.0 ~ 7.5 Kgm (Shop Manual page. 00-32)				
Oil cooler	New	Reuse																	
Front Side																			
Rear Side																			
Act :	kgm																		

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17	Common Rail		- Install Bracket (3) - Install common rail (2) - Common rail mounting bolts : Act : _____ kgm 4 ~ 5 kgm (Shop Manual Page. 50-26) * Install the common rail at this point since positioning of the high-pressure pipe needed when installing the fuel injector.																																														
18	Heat tab		- Install the water tube. - Water tube joint bolts : Act : _____ kgm 1.0 ~ 1.3 Kgm (Shop Manual Page. 50-28)																																														
			- Install heat label at cylinder head No.6 Act : _____ ° 99°C ~ 127°C (210°F ~ 260°F)																																														
21	Valve mechanis		Cross Heads - Fit cross heads and adjust so that when the screw has touched the valve stem it is advanced 20°. - Cross head adjusting screws go to exhaust manifold side of head. - Lock nut tighten torque: Act : _____ kgm																																														
			Push rod - Install Push rod in the hole * Check that the push rod is in the cam follower.																																														
	Valve mechanis		Fuel injector - Check that the inside of the injector Sleeve is free from dirt. *O-ring : coat with Engine oil *Ball washer: coat with Engine oil -Holder mounting bolt: Act = _____ kgm 6.0 ± 0.6Kgm (Shop Manual Page. 50-28) Write injectors code below: <table border="1"> <thead> <tr> <th>CYL.NO</th> <th>INJECTOR CODE</th> <th>HOLE OF NOZZLE</th> </tr> </thead> <tbody> <tr><td>1R</td><td></td><td></td></tr> <tr><td>2R</td><td></td><td></td></tr> <tr><td>3R</td><td></td><td></td></tr> <tr><td>4R</td><td></td><td></td></tr> <tr><td>5R</td><td></td><td></td></tr> <tr><td>6R</td><td></td><td></td></tr> <tr><td>1L</td><td></td><td></td></tr> <tr><td>2L</td><td></td><td></td></tr> <tr><td>3L</td><td></td><td></td></tr> <tr><td>4L</td><td></td><td></td></tr> <tr><td>5L</td><td></td><td></td></tr> <tr><td>6L</td><td></td><td></td></tr> </tbody> </table> *Example : Injector code 6290 05M 0128 Hole of nozzle: 8 Holes	CYL.NO	INJECTOR CODE	HOLE OF NOZZLE	1R			2R			3R			4R			5R			6R			1L			2L			3L			4L			5L			6L					CRITICAL ITEM CHECK <table border="1"> <tr> <td>Name :</td> <td>Name :</td> </tr> <tr> <td>QA INSPECTOR</td> <td>PROD. SUPERVISOR</td> </tr> </table>  3) Fit ball washer (10) to bolt (9) and then tighten injector holder (1) temporarily.  Ball washer: Engine oil (EO30) 	Name :	Name :	QA INSPECTOR	PROD. SUPERVISOR
CYL.NO	INJECTOR CODE	HOLE OF NOZZLE																																															
1R																																																	
2R																																																	
3R																																																	
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MAIN ASSEMBLY CHECK SHEET
ENGINE ASS'Y
KOMATSU ENGINE 12V140E-3

WO NUMBER

ENGINE MODEL

ENGINE S/N (new)

MACH.MODEL

SAA12V140-3

HD 785-7

DOCUMENT NO

RC/QA/EG/014

PUBLIC DATE

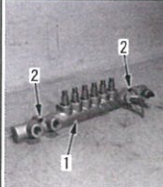
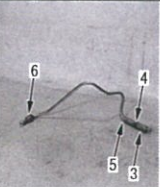
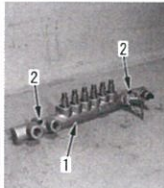
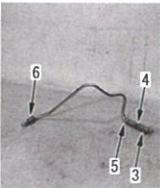
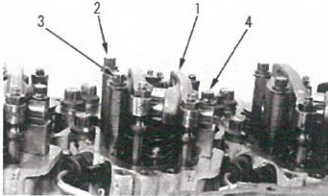
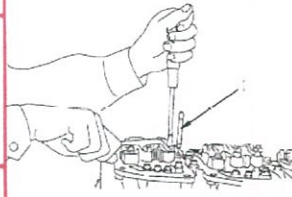
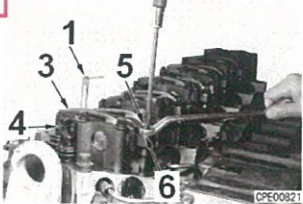
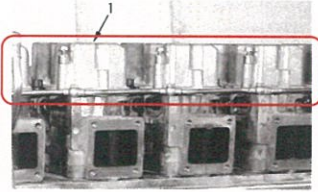
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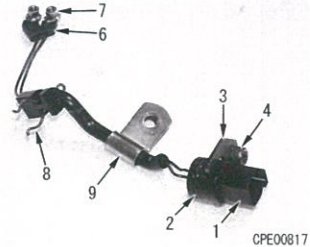
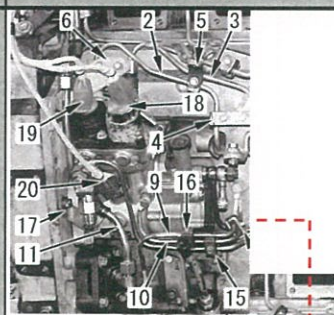
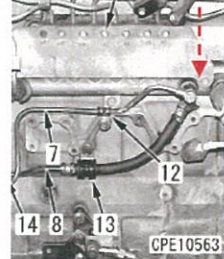
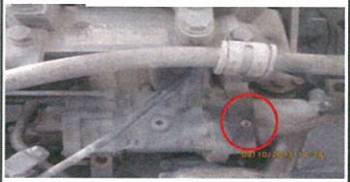
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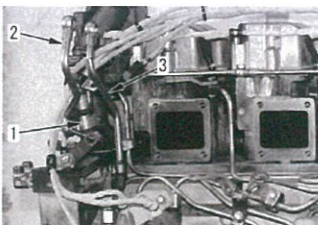
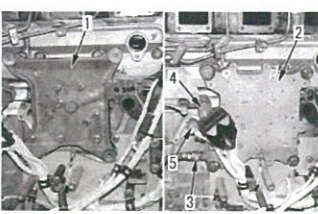
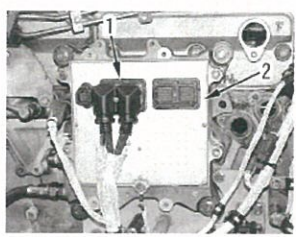
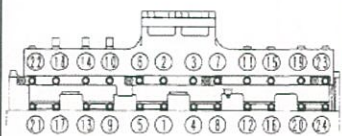
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		Common rail and high-pressure pipe	- Before installing the high-pressure pipe, check it for the following defects. If there is any of these defects, it can cause fuel leakage. Accordingly, replace the high-pressure pipe.					 
22	Valve Mechanis	Torque	- Tighten special bolts (2) : Act : kgm - Tighten Sleeve Nut (5)(6) : Act : kgm - Tighten Clamp : Act : kgm	6.02 ~ 7.55kgm (Shop Manual Page. 50-16) 4 ~ 5 kgm (Shop Manual Page. 50-17) 1.0 ± 0.1 kgm (Shop Manual Page. 50-17)				 
		Rocker Arm and Shaft Ass'y	-Install rocker arm and shaft assembly (1) and -tighten mounting bolts (2), (3) and (4). *Before tightening the mounting bolts, check that the ball of the adjustment screw is fitted to the socket of the push rod. - TightenTorque mounting bolts. Act : kgm	9.5 ~10.5 kgm (Shop Manual Page. 50-17)				
	Valve Mechanis	AdjustingV alve clarence	-Adjust valve clarence Intake : Act : mm -Adjust valve clarence. Exhaust : Act : mm -Tighten locknut (6) to secure adjustment screw (5) -Locknut: Act : kgm	0.35 ± 0.02 (Shop Manual Page. 13-39) 0.57 ± 0.02 (Shop Manual Page. 50-17) 5.4 ~6.6 kgm (Shop Manual Page. 50-18)	CRITICAL ITEM CHECK Name : QA INSPECTOR PROD. SUPERVISOR			
		Head cover	-Fit the O-ring and install cylinder head cover. -Since the bolt at the front on the air intake manifold side is also used to secure the wiring harness clamp, tighten it temporarily. -Mounting bolt: Act : kgm	3.0 ~ 3.5 kgm (Shop Manual Page. 50-18)				 

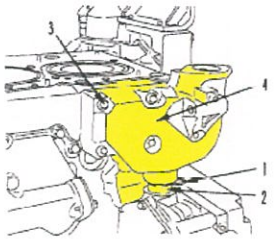
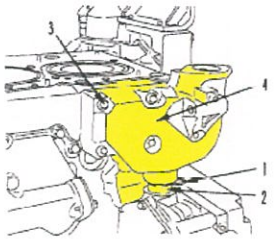
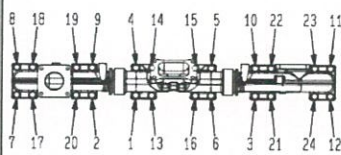
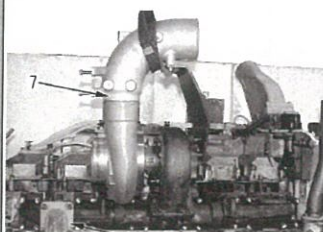
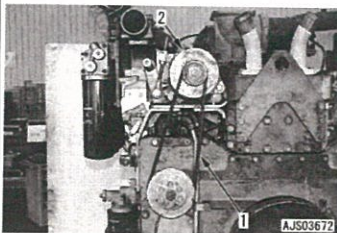
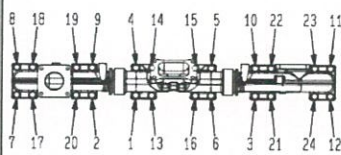
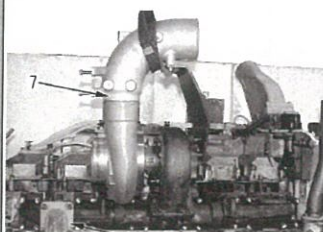
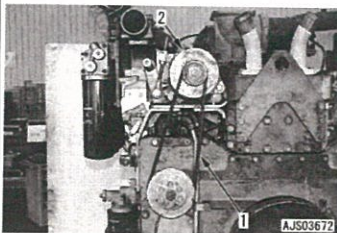
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23	Wiring Harness & Sensor Connector		- Install wiring harness terminal (6) to fuel injector (5) and tighten 2 nuts (7) - Terminal nut tighten torque: Act : kgm - Install sensor connector	0.2 ± 0.02 kgm (Shop Manual page. 50-22)				
No.	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN			SKETCH DRAWING or PHOTOS
					MEC	QA	SPV	
24	Piping & wiring of supply pump		- Install bracket (1). - install fuel tubes (2) and (3). Sleeve nut of tube : • Pump side Act : kgm • Common rail side Act : kgm - Clamping bolt : Act : kgm - Install fuel tubes (7) and (8). • Pump side Act : kgm • Common rail side Act : kgm - Install Oil tubes (9) (10) and (11). • Joint bolt of tubes (9) and (10). Act : kgm • Sleeve nut of tube (11) Act : kgm - Mounting bolt of clamps (14), (15) and (16): Act : kgm - Mounting supply pump(1): Act : kgm - Supply pump assy S/N: RH: LH:	4 ~ 5 kgm (Shop Manual page. 50-23) 4 ~ 6 kgm (Shop Manual page. 50-23) 1.0 ± 0.1 kgm (Shop Manual page. 50-23) 1.5 ~ 2.0 kgm (Shop Manual page. 50-23) 1.8 ~ 2.3 kgm (Shop Manual page. 50-23) 0.8 ~ 1.3 kgm (Shop Manual page. 50-23) 4.4 ~ 4.8 kgm (Shop Manual page. 50-23) 0.8 ~ 1.3 kgm (Shop Manual page. 50-23) 6.7 ± 0.7 kgm (Shop Manual page. 50-23)				  
25	Lock bolt	1	-Mounting Lock Bolt on Fuel pump drive RH and LH and install the washer gasket. Act: kgm (after finish install fuel pump assy to fuel Pump drive).	0.8 ~ 1.2 kgm (Shop Manual page. 50-23)				 

MAIN ASSEMBLY CHECK SHEET ENGINE ASS'Y KOMATSU ENGINE 12V140E-3	WO NUMBER		DOCUMENT NO	RC/QA/EG/014
	ENGINE MODEL	SAA12V140-3	PUBLIC DATE	28/08/2009
	ENGINE S/N (new)		REVISION NO.	06
	MACH.MODEL	HD 785-7	PAGE	10 / 12

26	Breather remote tube and corrosion resistor tube		- Install breather remote tube (3). - Install corrosion resistor tubes (2) and (1) temporarily					
27	Engine controller cooler		- Install the O-ring and plate (1). - Install engine controller cooler (2) and fuel hose (3) <i>*Install the hose on the engine controller cooler inlet side when installing the fuel filter head.</i> - Cooling plate mounting bolt: Act : kgm - Connect fuel pressure sensor connector PFUEL (4) and engine oil pressure sensor connector POIL (5)	3.5 ~ 4 kgm (Shop Manual page. 50-32)				
No.	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN			SKETCH DRAWING or PHOTOS
					MEC	QA	SPV	
28	Engine controller		- Install engine controller (2) Torque mounting bolt: Act : kgm And connect connector main harness(1) <u>-Engine controller S/N:</u> RH: LH:	2,4 kgm (Shop Manual page. 50-38)				
29	Air Intake Manifold		- Fit the gasket and sling and install air intake manifold (1) and then tighten the mounting bolts in the following order. - Mounting bolt: Act : kgm - Install air intake connector.	6.0 ~7.5 kgm (Shop Manual Page. 50-32)				

MAIN ASSEMBLY CHECK SHEET ENGINE ASS'Y KOMATSU ENGINE 12V140E-3	WO NUMBER		DOCUMENT NO	RC/QA/EG/014
	ENGINE MODEL	SAA12V140-3	PUBLIC.DATE	28/08/2009
	ENGINE S/N <i>(new)</i>		REVISION NO.	06
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30	Water Pump		-Fit O-ring and install water pump assembly <table><tr><td>Water Pump</td><td>New</td><td>Reuse</td></tr><tr><td></td><td></td><td></td></tr></table> * The mounting bolts of the water pump are used to tighten the gear case. Tightening Mounting bolt: <table><tr><td>Act :</td><td>kgm</td></tr></table> 9,5 ~ 10,5 Kgm (Shop Manual page. 00-32) <td></td> <td></td>	Water Pump	New	Reuse				Act :	kgm		
Water Pump	New	Reuse											
Act :	kgm												
31	Thermostat housing		-Fit O-ring to tube (21, then install to thermostat housing (4). -Fit gasket and install thermostat housing (4) with mounting bolts (3). -Move tube (2) down and install ring (1). Tightening Mounting bolt: <table><tr><td>Act :</td><td>kgm</td></tr></table> 6.7± 0.7 Kgm (Shop Manual page. 00-32) <td></td> <td></td>	Act :	kgm								
Act :	kgm												
32	Exhaust manifold & Turbo Charger	Exhaust manifold - Fit the gasket and sling and install exhaust manifold and then tighten the mounting bolts in the following order - Mounting bolt: <table><tr><td>Act :</td><td>kgm</td></tr></table> 6.0 ~7.5 Kgm (Shop Manual Page. 50-27) Turbo Charger - Fit the gasket and install turbocharger assembly - Turbo charger mounting bolt torque : <table><tr><td>Act :</td><td>kgm</td></tr></table> 4.5~5.0 Kgm (Shop Manual Page. 13-51) Alternator -Install alternator(2) -Install auxiliary equipment belt(1) -Adjust tension of auxiliary equipment belt. -adjust its deflection 20mmwhen the pushing force 6kgm,its applied to the center trough the finger. Adjuster lock bolt: <table><tr><td>Act :</td><td>kgm</td></tr></table> 5.5 ~ 12.5 Kgm S/N Alternator: <td></td> <td>    </td>	Act :	kgm	Act :	kgm	Act :	kgm		  			
Act :	kgm												
Act :	kgm												
Act :	kgm												
No.	PARTS	CHECK	CONDITIONS TO BE CHECKED	SPEC.	SIGN	SKETCH DRAWING or							

MAIN ASSEMBLY CHECK SHEET

ENGINE ASS'Y

KOMATSU ENGINE 12V140E-3

33 Starting Motor

Tightening Torque for Sensor

Sensor name	Tightening torque
NE	4.9 – 7.9 Nm {0.5 – 0.8 kgm}
Engine speed	69 – 74 Nm{7 – 7.5 kgm} (Locknut)
Fuel temperature	23 ± 3 Nm {2.35 ± 0.3 kgm}
Boost pressure	23 ± 3 Nm {2.35 ± 0.3 kgm}
Intake air temperature	23 ± 3 Nm {2.35 ± 0.3 kgm}
Coolant temperature	23 ± 3 Nm {2.35 ± 0.3 kgm}
EGR gas pressure	23 ± 3 Nm {2.35 ± 0.3 kgm}
Engine oil pressure	23 ± 3 Nm {2.35 ± 0.3 kgm}
Atmospheric pressure	7.3 – 12.7 Nm {0.7 – 1.3 kgm} (Hexagon head bolt) 4.1 – 9.5 Nm {0.4 – 1.0 kgm} (Hexagon socket head bolt)
Fuel pressure	Never disassemble.

Notes :

ASSEMBLY BY

INSPECTED & APPROVED BYMECHANIC (MEC)

QUALITY ASSURANCE (QA)

GL/SPV PRODUCTION



PT SAPTAINDRA SEJATI

ENGINE ASSY

ENGINE PERFORMANCE TEST

CHECKSHEET

KOMATSU ENGINE 12V140E-3 SERIES

MACHINE MODEL

HD785-7

RO NO :

ENGINE SECTION

DOCUMENT NO.	RC/QR05/EG/02/01
PUBLICATION DATE	18/08/2009
REVISION NO.	01
PAGE	

WO NUMBER		START DATE / TIME			
ENGINE MODEL	SAA12V140E-3	FINISH DATE / TIME		Name :	Name :
ENGINE SER.NO.		DYNO.ARM LENGTH	716 mm	QA INSPECTOR	PROD. SUPERVISOR
MACHINE MODEL	HD 785-7	TEST BY			

No	STANDARD (MANUAL)		ACTUAL TESTED	
	RPM	HP	RPM	HP
1	650	0		
2	1000	79		
3	1250	197		
4	1550	489		
5	1800	852		
6	1900	1200		

STANDARD PRESSURE		ACTUAL		REMARK
MAXIMUM PRESSURE	300 mmH ₂ O(2.94 Kpa)	1st		
SERVICE LIMIT VALUE	600 mmH ₂ O(5.88 Kpa)	2nd		
ALTERNATOR OUTPUT VOLTAGE	STANDARD	ACTUAL		REMARK
	27.5 ~ 29.5			

SPEED (Rpm)		OIL		WATER		BOOST PRESS (mmHg)	BLOW BY (mmH2O)	EXH TEMP (°C)
		Press. (Kg/cm²)	Temp (°C)	Press. (Kg/cm²)	Temp (°C)			
Low idle	650 ± 20	Min. 0.8	90 ~ 110	-	70~90	Min. 1250 at rated HP	Max 300 permissible 600	Max 650 at rated HP and Max. 700 at Max.torque
High idle	2250 ± 25	3.0 ~ 4.5						
POWER (HP) / RPM		1160 ~ 1240 / 1900 ± 50						
TORQUE (Kgm) / RPM		502~ 534 / 1350 ± 50						
EXHAUST TEMP DIFF		Maximum 10°C RH and LH Bank						

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[illegible]

ENGINE PERFORMANCE TEST REPORT	DOCUMENT NO.	RC/QR05/EG/02/01	
	PUBLICATION DATE	18/08/2009	
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	PAGE		

WO NUMBER		START DATE / TIME	
ENGINE MODEL	SAA12V140E-3	FINISH DATE / TIME	
ENGINE SER.NO.		DYNO.ARM LENGTH	716 mm
MACHINE MODEL	HD 785-7	TEST BY	

CRITICAL ITEM CHECK

Name : Name :

QA INSPECTOR PROD. SUPERVISOR

RECHECK VALVE CLEARANCE AFTER LOAD

Adjustment valve clearance 	* Penentuan posisi top cylinder engine harus dicek dan disetujui oleh QA. Step and method adjustment of value clearance following to the shop manual attached. Firing Order: R1-L1-R5-L5-R3-L3-R6-L6-R2-L2-R4-L4. Valve clearance:								
	Intake valve : 0.35 ± 0.02	Act : mm	<table border="1"> <thead> <tr> <th colspan="2">PARAF</th> </tr> <tr> <th>MEC</th> <th>QA</th> </tr> </thead> <tbody> <tr> <td></td> <td></td> </tr> </tbody> </table>	PARAF		MEC	QA		
	PARAF								
	MEC	QA							
Exhaust valve : 0.57 ± 0.02	Act : mm								
Tighten locknut to secure adjustment screw. Locknut: 5.4 ~ 6.6 kgm	Act : kgm								
Rocker Arm cover 	Fit O ring and install rocker arm cover. Since the bolt at the front on the air intake manifold side is also used to secure the wiring harness clamp, tighten it temporary.								
	Mounting bolt : 3.0 ~ 3.5 kgm.	Act : kgm							

INSPECTION BY

INSPECTION & APPROVAL BY

MECHANIC

QUALITY ASSURANCE (QA)

GL/SPV PRODUCTION

DELIVERY INSPECTION SHEET ENGINE

MACHINE MODEL	HD785-7	ENGINE PART NO.		DOCUMENT NO.	RC/QR01/EG/01/01
ENGINE MODEL	SAA12V140E-3	WO ORDER NO.		PUBL. DATE	
ENGINE NO.		RECEIVING DATE		REVISION NO.	01
CUSTOMER NAME				PAGE	2 OF 2

BELOW ARE THE ITEMS RELATED TO DELIVERY INSPECTION AND WHILE YOU DOING THIS INSPECTION JOB, PLEASE REFERRING TO THE REMANUFACTURED COMPONENTS - STANDARD CONFIGURATION.

✓ : GOOD CONDITION

X : BAD CONDITION

⊗ : NONE

SKETCH DRAWING OR PHOTOS

NO.

ITEM

INSPECTION
CONDITION

Right Side View

A photograph showing the right side of a large industrial engine. Red arrows point from numbered circles (42-60) to various components. 42 points to the air connector, 43 to the exhaust brake, 44 to the heater switch, 45 to the air intake manifold, 46 to the oil cooler, 47 to the cooling system piping, 48 to the fuel pressure sensor, 49 to the common rail, 50 to the priming pump, 51 to the fuel supply pump, 52 to the oil pan, 53 to the water inlet cover, 54 to the water pump, 55 to the fuel filter, 56 to the alternator safety guard, 57 to the breather tube, 58 to the oil filter, 59 to the alternator, and 60 to the air intake connection.

Front Side View

A photograph showing the front side of the engine, focusing on the cooling fan and pulley system. Red arrows point from numbered circles (62-78) to various components. 62 points to the front hanger plate, 63 to the fan drive pulley, 64 to the fuel return block, 65 to the V-belt alternator, 66 to the alternator drive pulley, 67 to the tension pulley, 68 to the vibration damper, 69 to the front support, 70 to the crank pulley, 71 to the pointer, 72 to the yoke, 73 to the V-belt fan set, 74 to the blowby sensor, 75 to the water temperature sensor, 76 to the thermostat housing, 77 to the water outlet tube, and 78 to the cooling fan.

42 Air Connector RH

43 Exhaust Brake Assy RH

44 Heater Switch Assy With Cover RH

45 Air Intake Manifold RH

46 Oil Cooler Assy

47 Tube Cooling System Piping

48 Fuel Pressure Sensor With Limiter Assy RH

49 Common Rail Assy

50 Priming Pump Assy With Cover RH

51 Fuel Supply Pump Assy RH (P/N : 6219-71-1120)

S/N :

52 Oil Pan

53 Water Inlet Cover Plate

54 Water Pump Assy

55 Fuel Filter Assy

56 Cover Safety Guard Alternator

57 Tube, Breather With Hose RH

58 Oil Filter Assy

59 Alternator Assy (90A) (P/N : 600-825-9330)

S/N :

60 Tube Air Intake Connection RH

62 Plate Front Hanger

63 Fan Drive With Pulley Assy

64 Block Fuel Return With Sensor Assy RH & LH

65 V-Belt Alternator

66 Alternator Drive Pulley

67 Tension Pulley Assy

68 Vibration Damper Assy (P/N : 6215-31-8200)

69 Front Support

70 Crank Pulley Assy

71 Pointer

72 Yoke

73 V-Belt Fan Set

74 Blowby Sensor Assy

75 Water Temperature Sensor

76 Thermostat Housing

77 Tube Water Outlet (2 pcs)

78 Cooling Fan

Notes :

Inspected by,

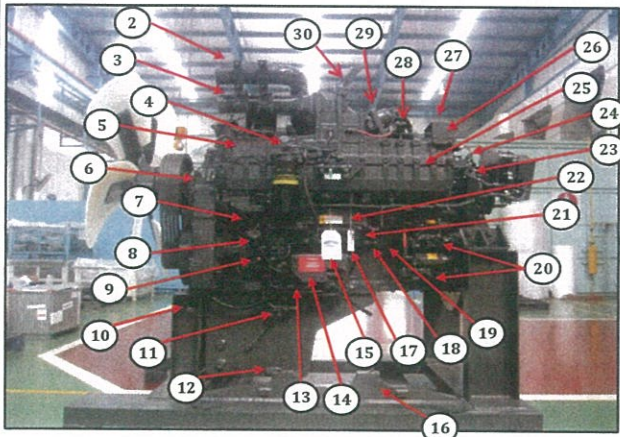
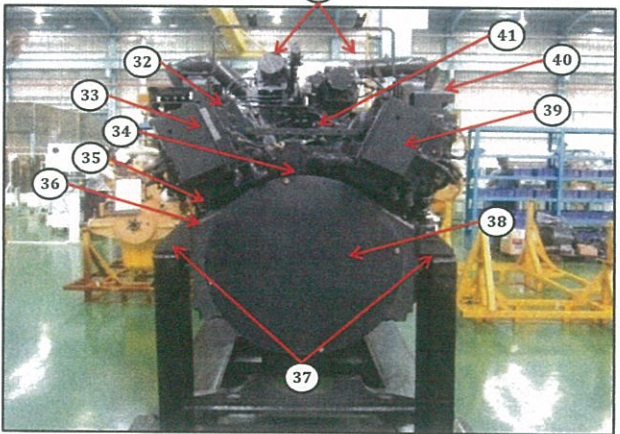
Aproved by,

DELIVERY
INSPECTION SHEET
ENGINE

MACHINE MODEL	HD785-7	ENGINE PART NO.		DOCUMENT NO.	RC/QR01/EG/01/01
ENGINE MODEL	SAA12V140E-3	WO ORDER NO.		PUBL. DATE	
ENGINE NO.		RECEIVING DATE		REVISION NO.	01
CUSTOMER NAME				PAGE	1 OF 2

BELOW ARE THE ITEMS RELATED TO DELIVERY INSPECTION AND WHILE YOU DOING THIS INSPECTION JOB, PLEASE REFERRING TO THE REMANUFACTURED COMPONENTS - STANDARD CONFIGURATION.

✓ : GOOD CONDITION X : BAD CONDITION ⊗ : NONE

SKETCH DRAWING OR PHOTOES	NO.	ITEM	INSPECTION CONDITION
Left Side View 	1	Painting (Y/B)	
	2	Tube Air Intake Connection LH	
	3	Air Connector LH	
	4	Fuel Filter Assy LH	
	5	Engine Name Plate	
	6	Tube, Breather With Hose LH	
	7	Fuel Supply Pump Assy LH (P/N : 6219-71-1110) S/N :	
	8	Tube Oil Filler Cap	
	9	Gauge Oil (STD : H+ 0~10 mm) Act : H +mm	
	10	Engine Stand (Original / Std) Stand No :	
	11	Oil Level Sensor Assy	
	12	Drain Oil Plug	
	13	Priming Pump Assy With Cover LH	
	14	Warning Sticker Oil Level	
	15	Bracket Mounting	
	16	Quality passed Inspection Tag	
	17	Caution Tag	
	18	Oil Temperature Sensor	
	19	Oil Pressure Sensor	
	20	Starting Motor Assy (7.5 KW) (P/N : 600-813-7542) Upper S/N : Lower S/N :	
	21	Fuel Pressure Sensor With Limiter Assy LH	
	22	Common Rail Assy	
	23	Boost Pressure Sensor	
	24	Intake Temperature Sensor	
	25	Air Intake Manifold LH	
	26	Exhaust Temperature Sensor (4 pcs) With Bracket	
	27	Customer Responsibilities Sticker	
	28	Heater Switch Assy With Cover LH	
	29	Exhaust Brake Assy LH	
	30	Bracket Exhaust Brake Mounting	
Rear Side View 	31	Turbocharger Assy RH P/N : 6505-67-5030 S/N :	
		Turbocharger Assy LH P/N : 6505-67-5040 S/N :	
	32	Heat Tab position	
	33	Engine Controller Assy LH S/N : ESN : P/N Assy :	
	34	Caution Flywheel Sticker	
	35	Pick Up Revolution Sensor	
	36	Flywheel Housing	
	37	Rear Bracket Mounting (2 pcs)	
	38	Flywheel (P/N : 6219-31-4110)	
	39	Engine Controller Assy RH S/N : ESN : P/N Assy :	
	40	Stiker Attention Installation Report	
41	Exhaust Manifold Assy RH & LH		

Notes :